Cover Page

Standard: BAC5004-2 Rev: (AG) 07-May-2015

WARNING - The Export Control or Intellectual Property language on this page takes precedence over any such language that may appear on subsequent pages of this document.

No license is required for the dissemination of the commercial information contained herein to foreign persons other than those from or in the terrorist supporting countries identified in the United States Export Administration Regulations (EAR) (15 CFR 730-774). It is the responsibility of the individual in control of this data to abide by U.S. export laws.

ECCN 9E991

Boeing Proprietary, Distribution limited to Boeing employees and others covered by applicable proprietary information or nondisclosure agreements.

Copyright © 1992, 1996, 2003, 2006, 2008, 2010-2012, 2015 Boeing. All rights reserved.

Any reproduction of this document or portions thereof must also reproduce this page of the document.

1 SCOPE

NOTE: Incorporated PSDs: None
Cancelled PSDs: None

a. This specification and BAC5004 establishes the requirements for installation and inspection of non-fluid tight, permanent type, straight shank fasteners in metal structure as specified in the applicable Engineering drawings. For this specification, the following are considered as permanent fasteners: Hi-Shear rivets, lockbolts and hex-drive fasteners.

b. The Engineering requirements for installation and inspection of fasteners in fiber/resin composites are found in BAC5063. BAC5063 also provides requirements for fasteners installed in stack-ups containing any fiber/resin composite and thermoplastic material.

c. The Engineering requirements for installation and inspection of fasteners in fluid tight applications are found in BAC5047.

d. Refer to BAC001PREF for guidance on use of Boeing process specifications and Boeing process specification departures.

WARNING WARNINGs may be included throughout this specification. Do not take these WARNINGs to be all inclusive, nor to completely describe hazards or precautionary measures applicable to specific procedures or operating environments.
Non-Boeing personnel must refer to their employer's safety instructions for information concerning hazards which may occur during operations described in this specification.

2 CLASSIFICATION

Not applicable to this specification.

3 REFERENCES

The current issue of the following documents shall be considered a part of this specification to the extent herein indicated.

ANSI/ASME B18.3	-	Socket Cap, Shoulder, Set Screws, and Hex Keys (Inch Series)
ANSI/ASME B107.17M	-	Gages and Mandrels for Wrench Openings
AS9100	-	Quality Management Systems - Requirements for Aviation, Space and Defense Organizations
BAC5004	-	Installation of Permanent Fasteners
BAC5047	-	Fluid Tight Fastener Installation
BAC5063	-	Fastener Installation in Composite Structures
BAC5300	-	Forming, Straightening and Fitting Metal Parts
BAC5408	-	Vapor Degreasing
BAC5492	-	Machining and Cutting Titanium
BAC5504	-	Integral Fuel Tank Structure Sealing
BSS7083	-	Torque Procedures and Tools for Fastener Installation
BSS7286	-	Statistical Process Control of Designated Engineering Characteristics

Copyright © Boeing. All rights reserved. See Cover Page.

3 **REFERENCES (Continued)**

D6-56617	-	Certification of Automatic Electric and Hydraulic-Squeeze Fastening Equipment
FED-STD-H28	-	Screw Thread Standards for Federal Services

Copyright © Boeing. All rights reserved. See Cover Page.

4 CONTENTS

Copyright © Boeing. All rights reserved. See Cover Page.

4 CONTENTS (Continued)

Copyright © Boeing. All rights reserved. See Cover Page.

4 **CONTENTS (Continued)**

LIST OF FIGURES

Copyright © Boeing. All rights reserved. See Cover Page.

4 **CONTENTS (Continued)**

LIST OF TABLES

Copyright © Boeing. All rights reserved. See Cover Page.

4 CONTENTS (Continued)

Copyright © Boeing. All rights reserved. See Cover Page.

5 MATERIALS CONTROL

Not applicable to this specification.

6 FACILITIES CONTROL

6.1 SWAGE PRESSURE

a. Automatic fastener installation machines shall be equipped with pressure setting controls which regulate collar swage load. The control settings shall be related to the required collar swage on a machine-by-machine basis.

b. Manual swage tools shall be equipped with either pressure setting controls or positional control of the swage stroke.

6.2 COUPON ASSEMBLY

For automatic fastening machines, a set-up coupon assembly shall be prepared for each production assembly, prior to panel fastening, to verify machine function and settings. All applicable requirements shall be met by the coupon, including hole quality, collar swage, collar/nut gapping, head seating, pin protrusion and sealant application (If applicable). Each fastener diameter shall be verified by the coupon, or coupons.

7 DEFINITIONS

The following definitions apply to terms that are uncommon or have special meaning as used in this specification.

Automatic Fastening Machine - Machine capable of drilling (and countersinking, if necessary), inserting and seating bolts and installing collars, in one clamped condition.

Inspection Reliability Requirement - The minimum probability of product conformance that the inspection procedure allows. This is an index used in defining reliability-based sampling plans.

Key Characteristics (KC) - A measurable feature whose variation has the greatest impact on the fit, form, performance, and service life of the finished part from the perspective of the customer. This feature may be measured directly or indirectly through a secondary set of process parameters.

Key Process Parameters (KPP) - Process parameters that contribute to variation of a key characteristic. These are most effectively determined by the use of a designed experiment.

ST Tool - Standard Tool available from the Boeing Company.

Copyright © Boeing. All rights reserved. See Cover Page.

8 MANUFACTURING CONTROL

WARNING

This specification involves the use of chemical substances which are hazardous. Boeing personnel shall refer to the work area Hazard Communication Handbook for health effect and control measure information contained in the HazCom Info Sheets and Material Safety Data Sheets. For disposition of hazardous waste materials, consult site environmental engineers for proper disposal methods.

Non-Boeing personnel should refer to manufacturer's Material Safety Data Sheet(s) and their employer's safety instructions.

The following items apply to all permanent fasteners according to this specification:

a. Position and fit parts to meet the requirements of the Engineering drawing and BAC5300 prior to fastener installation.

b. All dimensions in this specification are in inches.

c. See BAC5492 for restrictions regarding tooling in contact with titanium.

8.1 HOLES

a. Parts shall be firmly clamped together prior to and during the drilling, reaming, and fastening processes. Method of clamping shall not mark or damage the finish or the structure.

b. Hole diameter limits shall be in accordance with Table I, unless otherwise specified on the drawing. Transition fit hole diameters are applicable to aluminum structures, and close ream hole diameters are applicable to titanium and steel structures, and combinations with aluminum structure.

c. Maximum hole angularity and countersink angularity shall be as shown in Figure 1. When countersinking in accordance with the requirements of Figure 1, compliance with the end item installation requirements in Section 11.2.1 shall be considered.

NOTE: To minimize countersink eccentricity, an integral drill/countersink tool or piloted countersink tool is recommended.

Copyright © Boeing. All rights reserved. See Cover Page.

8.1 HOLES (Continued)

TABLE I - HOLE SIZES

NOMINAL FASTENER SHANK DIAMETER	HOLE DIAMETER LIMITS FOR ALUMINUM STRUCTURES (INCH) (TRANSITION FIT)		HOLE DIAMETER LIMITS FOR TITANIUM, NICKEL ALLOY AND STEEL STRUCTURES, AND COMBINATIONS WITH ALUMINUM (INCH) (CLOSE REAM)	
	MIN.	MAX.	MIN.	MAX.
0.1640	0.161 FL 1	0.164 FL 1	0.1635	0.1645
0.1900	0.187	0.190	0.1895	0.1905
0.2500	0.247	0.250	0.2495	0.2505
0.3125	0.309	0.313	0.3120	0.3130
0.3750	0.371	0.375	0.3745	0.3755
0.4375	0.434	0.438	0.4370	0.4380
0.5000	0.496	0.500	0.4995	0.5005
0.5625	0.559	0.563	0.5620	0.5630
0.6250	0.621	0.625	0.6245	0.6255
0.7500	0.746	0.750	0.7495	0.7505
0.8750	0.871	0.875	0.8745	0.8755
1.0000	0.996	1.000	0.9995	1.0005

FL 1 For 0.164 inch diameter BACB30DX, BACB30DY, BACB30GP, and BACB30GQ the hole diameter limits are 0.162 to 0.165 inch.

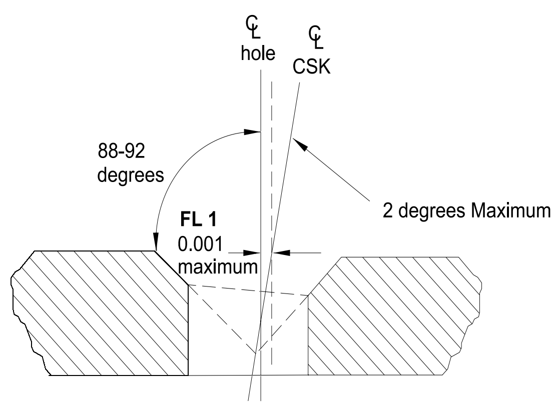


FIGURE 1 - LIMITS FOR COUNTERSINK ANGULARITY AND ECCENTRICITY AND HOLE ANGULARITY

FL 1 Hole to countersink centerline eccentricity measured at surface of structure.

d. A head-to-shank fillet relief is required for all fastener holes in all metals except as noted below. The head-to-shank fillet relief shall be provided by a radius or chamfer in the hole, or by the use of a countersunk washer. See Section 8.2.4 for washer use. Head-to-shank fillet relief provided by a radius or chamfer in the hole shall be in accordance with Table II and Figure 2 or Figure 3.

Copyright © Boeing. All rights reserved. See Cover Page.

8.1 HOLES (Continued)

Exception: Protruding head fasteners 0.375 inch and smaller nominal diameter may be hand or machine installed in aluminum or magnesium structure in noncold worked transition fit or interference fit holes without head-to-shank fillet relief, provided the head gap requirements of Section 11.2.1 are met.

NOTE: The use of fillet relief washers is not encouraged, because they increase weight. They may be used when it is not practical to radius or chamfer the holes.

TABLE II - RADIUS OR CHAMFER LIMITS FOR FASTENER HEAD-TO-SHANK FILLET RELIEF

NOMINAL SHANK DIA. (INCH)	FLUSH HEAD FASTENERS R (INCH)		PROTRUDING HEAD FASTENERS							
			ALL MATERIALS R (INCH)		IN ALL ALUMINUM/MAGNESIUM, INCLUDING MACHINE INSTALLED OR COLDWORKED HOLES			IN ALL MATERIALS EXCEPT ALUMINUM AND MAGNESIUM		
					D CHAMFER DIA. (INCH) FL 1 FL 2		MINIMUM SHEET THICKNESS FOR A RADIUS OR CHAMFER (INCH)	D CHAMFER DIA (INCH) FL 1		MINIMUM SHEET THICKNESS FOR A RADIUS OR CHAMFER (INCH)
	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.		MAX.	MIN.	
0.1640	0.040	0.030	0.035	0.025	0.208	0.188	0.060	0.233	0.218	0.107
0.1900	0.040	0.030	0.035	0.025	0.234	0.214	0.060	0.257	0.242	0.127
0.2500	0.040	0.030	0.035	0.025	0.294	0.274	0.060	0.320	0.300	0.150
0.3125	0.050	0.040	0.040	0.030	0.363	0.343	0.060	0.392	0.372	0.183
0.3750	0.050	0.040	0.040	0.030	0.426	0.406	0.060	0.455	0.435	0.209
0.4375	0.060	0.050	0.040	0.030	0.488	0.468	0.060	0.517	0.497	0.236
0.5000	0.060	0.050	0.040	0.030	0.551	0.531	0.060	0.580	0.560	0.253
0.5625	0.060	0.050	0.050	0.040	0.626	0.606	0.075	0.677	0.657	0.320
0.6250	0.060	0.050	0.050	0.040	0.688	0.668	0.075	0.739	0.719	0.320
0.7500	0.060	0.050	0.055	0.045	0.820	0.800	0.085	0.876	0.856	0.353
0.8750	0.060	0.050	0.060	0.050	0.951	0.931	0.090	1.013	0.993	0.387
1.0000	0.060	0.050	0.070	0.060	1.089	1.069	0.105	1.162	1.142	0.453

FL 1 For 0.0156 inch oversize, add 0.016 inch to both the max. and min. diameters. For 0.0312 inch oversize, add 0.031 inch to both the max. and min. diameters.

FL 2 Fasteners 0.375 inch and smaller in noncoldworked interference fit or transition fit holes may be installed without chamfer, provided the bolts are driven to seat the head and the gap requirements of Section 11.2.1 are met. See Section 8.1.d.

e. Unless otherwise specified, the countersink included angle for flush head fasteners is 99 to 101 degrees. See Figure 3.

f. For countersunk holes, the hole diameter is not required to meet the hole diameter tolerance when the remaining wall depth is 10 percent or less of the sheet thickness. Insure the hole diameter is in tolerance prior to countersinking.

Copyright © Boeing. All rights reserved. See Cover Page.

8.1 HOLES (Continued)

g. See BAC5004 (basic) for additional hole requirements. For designated sonic areas, break hole-edges in accordance with BAC5300-1.

EXCEPTION: Fastener holes in sonic areas do not require edge break when fasteners are installed with transition fit or interference fit.

h. An exit burr of 0.005 inch maximum is permissible from the hole when the fastener is installed with automatic fastening equipment.

i. For flush head bolts, a fillet relief radius is required at the juncture of the countersink and the hole as shown in Figure 3.

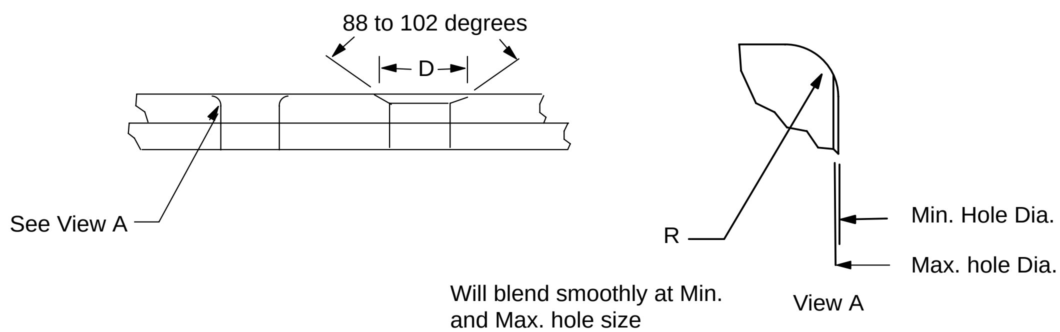


FIGURE 2 - HOLE RADII AND CHAMFER FOR PROTRUDING HEAD FASTENER

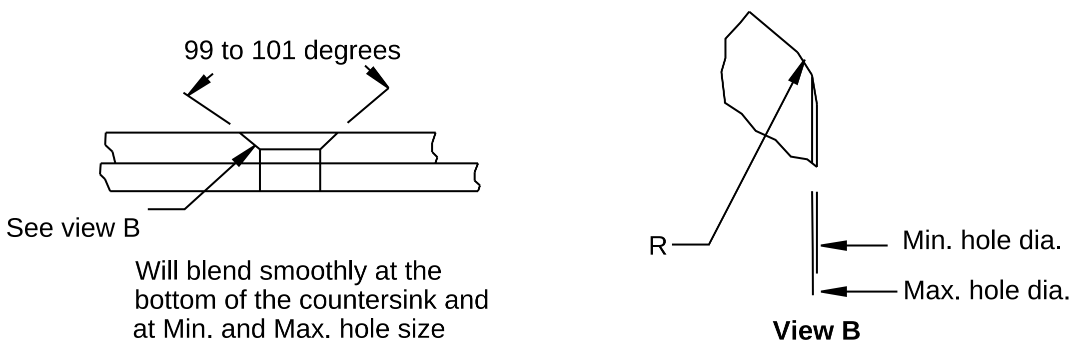


FIGURE 3 - HOLE AND COUNTERSINK FOR FLUSH HEAD FASTENER

8.2 INSTALLATION

8.2.1 FASTENER INSTALLATION

a. All fasteners installed in transition or interference fit holes shall have the manufactured head fully seated to the structure prior to collar installation. Except as noted in Section 8.2.8, the seating will be from the head side. Joints should be firmly clamped at the time of bolt installation. For fasteners installed in close-ream or clearance fit holes, fillet relief per Section 8.1.d. shall be applied prior to collar installation.

b. Shaving of the heads of flush head fasteners to meet aerodynamic flushness is not permitted.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.1 FASTENER INSTALLATION (Continued)

c. Unless otherwise approved by Engineering drawing, the fasteners shall not be reworked, degreased or have additional lubrication added.

d. Installed fasteners shall meet the protrusion requirements of Section 11.3. The protrusion requirements will be met when the appropriate grip length fastener is installed. Appropriate grip length may be determined using a standard grip scale (available from Omega Technologies, Canoga Park, CA).

(1) For callouts which do not specify washers, the appropriate grip length fastener is that which provides the required protrusion without washers. For callouts which specify washers, the appropriate grip length fastener is that which provides the required protrusion with the specified washers, except when counterbored nuts are used.

(2) When counterbored nuts are used, the appropriate grip length fastener is that which provides the required protrusion without the counterbore substitution washers. See Section 8.2.9 for proper use of washers with non-counterbored nuts.

e. Manufacturing tolerances on the heads of flush-head fasteners may result in partially exposed countersinks (see Figure 4). Exposed areas around the fastener head are acceptable if fastener tolerances, hole tolerances, flushness requirements, and head gapping requirements are met.

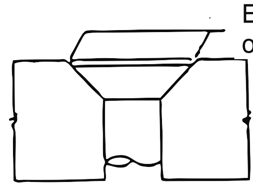


FIGURE 4 - PARTIALLY EXPOSED COUNTERSINK

8.2.2 GRIP LENGTH ADJUSTMENT

a. Grip length adjustment is necessary when fastener grip length specified does not provide the required protrusion. Fastener grip length adjustment may be made without drawing change. Appropriate grip length may be determined using grip scale 2-612. (Available from Omega Technologies, Canoga Park, CA).

b. Grip length adjustment will be necessary when the appropriate grip length fastener is not available (see Section 8.2.1.d.). Longer grip length fasteners may be used and washers shall be used to obtain the required protrusion. See Section 8.2.4 for washer use and selection.

c. Grip length adjustment may be necessary when washers are required for fillet relief. Where washers are required the fastener grip length adjustment may be made without drawing change. See Section 8.2.4 for washer use and selection.

d. See Section 8.2.4.b.(5) for limitations when using washers for grip length adjustment.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.3 CORROSION PROTECTION

All bare holes, countersinks, or spotfaces in magnesium shall be protected in accordance with the finish requirements on the Engineering drawing.

8.2.4 USE OF WASHERS

Washers may be used for grip length adjustment, counterbore substitution and fillet relief. Washer use shall comply with the following limitations. See Figure 8 for summary of use and limitations.

a. Fillet relief washers are permitted by Section 8.1.d.

(1) Only one washer may be used for fillet relief.

(2) Washer material shall be compatible with structure material. Washer material shall be determined from Table III.

(3) Washer part number shall be determined from Table IV.

(4) Only countersunk washers shall be used. Locate countersink under head of fastener.

b. Grip adjustment washers are permitted by Section 8.2.2.b.

(1) Washer material shall be compatible with both structure and collar material. Washer material shall be determined from Table III.

(2) Washer part numbers shall be determined from Table IV.

(3) Grip adjustment washers may be used under the head of the fastener as follows:

(a) When a grip adjustment washer is used directly under the head of the fastener, a countersunk washer shall be used and the countersink shall be under the head of the bolt (see Figure 5).

(b) Countersunk grip adjustment washers may also serve as a fillet relief washer (see Figure 5).

(4) Grip adjustment washers may be used under the fastener collar or under the fastener nut as follows:

(a) Plain (non-countersunk) washers shall be used.

(b) Grip adjustment washer(s) shall not be used under self sealing collars (BACC30BP) or self sealing nuts (BACN10TN, BACN10WM and BACN11BT).

(c) Grip adjustment washer(s) shall not be used under the self-aligning concave washer base of self-aligning washers (BACW10CA) (see Figure 5).

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.4 USE OF WASHERS (Continued)

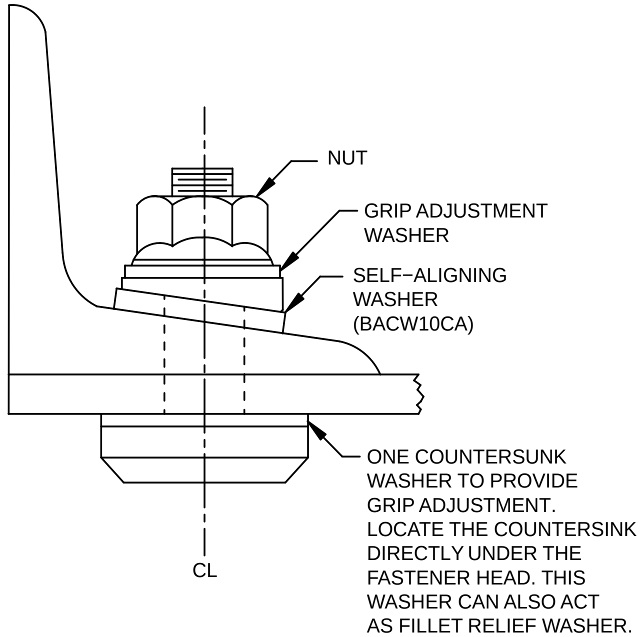


FIGURE 5 - USE OF WASHERS WITH SELF-ALIGNING WASHERS

(d) Grip adjustment washer(s) may be located between the collar or nut and the self aligning washers (BACW10CA) as shown in Figure 5.

(e) Following grip adjustment washer(s) may be used under the counterbore substitution washer(s) of self-aligning nut/washer combinations (BACN10MT / BACW10AU): (See Figure 6)

0.063 thick aluminum (anodized 7075 Alloy):

- BACW10P416AN through BACW10P421AN

Cadmium plated CRES steel:

- 0.032 thick: BACW10P410CG through BACW10P415CG.
- 0.063 thick: BACW10P416CG through BACW10P421CG.

BACW10P shall not interfere with the adjacent structure. In cases where there is interference, contact Engineering Liaison. Radius Fillers or Radiused Washers (as allowed by Engineering Liaison) shall be installed in a manner that does not allow the corners to rotate and ride into the structure during installation of the fastener. See Figure 7.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.4 USE OF WASHERS (Continued)

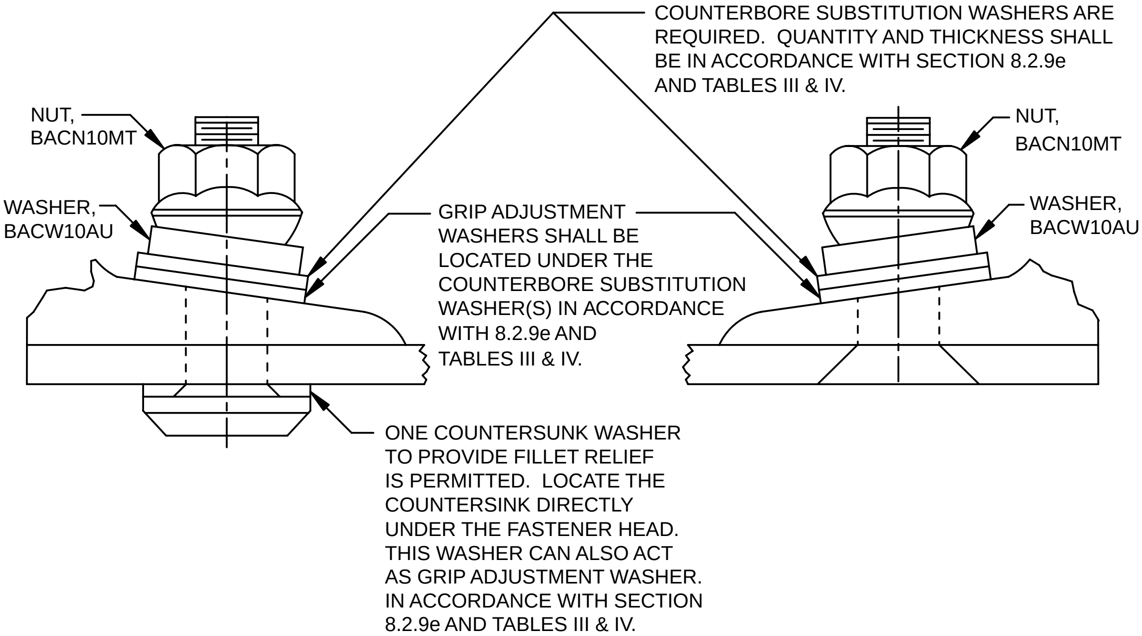


FIGURE 6 - USE OF WASHER WITH SELF-ALIGNING NUTS

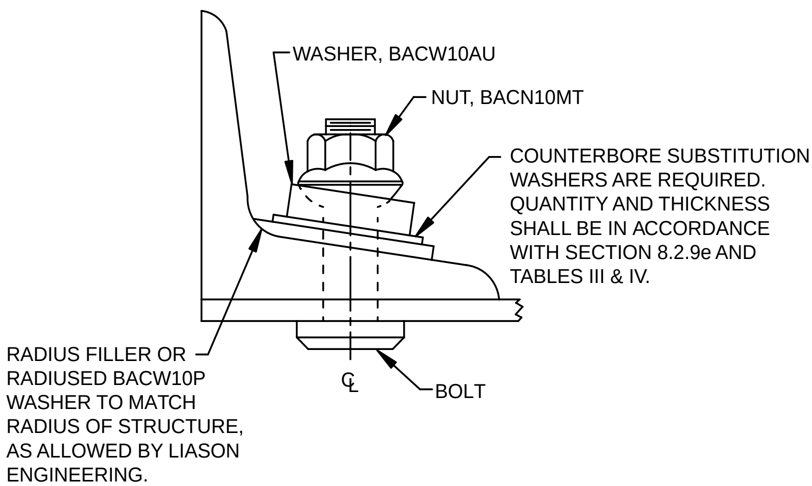


FIGURE 7 - USE OF RADIUS FILLER OR RADIUSED WASHERS WITH SELF-ALIGNING NUTS

(f) 0.032 and 0.063 thick aluminum alloy washers, BACW10AT2( )AS and BACW10AT3( )AS, may be used for grip adjustment under the self-aligning concave washer base of self-aligning collar / washer combinations (BACC30BQ or BACC30AG)

(g) Grip adjustment washers are in addition to the nut counterbore substitution washers (see Figure 13).

(5) Maximum number of grip adjustment washers used is two with the maximum total thickness available for the washers as shown in Table IV.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.4 USE OF WASHERS (Continued)

c. Lockwashers shall not be used.

d. See Figure 8 for washer use summary of limitations for steel, aluminum, or titanium structure.

e. Counterbore substitution washers are required by Section 8.2.9.e.

(1) Washer material shall be compatible with both structure and nut material. Washer material shall be determined from Table III.

(2) Washer part numbers shall be determined from Table IV.

(3) Only plain (noncountersunk) washers shall be used.

(4) Locate counterbore substitution washers in accordance with Figure 13.

f. For self-aligning nut/washer combination (BACN10MT and BACW10AU), see Figure 7. A self-aligning washer is always located adjacent to structure. When a BACW10CA OR BACW10EK self aligning washer is required by Engineering drawing, the fastener head shall be located on the non-sloping side of the hole, with the BACW10CA OR BACW10EK washer installed under the collar or nut unless otherwise specified by Engineering drawing. Use them on hex-drive bolts under the collar or nut for slopes up to 8 degrees. See Figure 5.

g. Washers shall not be used between seal nuts or seal collars of hex drive fasteners, unless permitted by Engineering drawing.

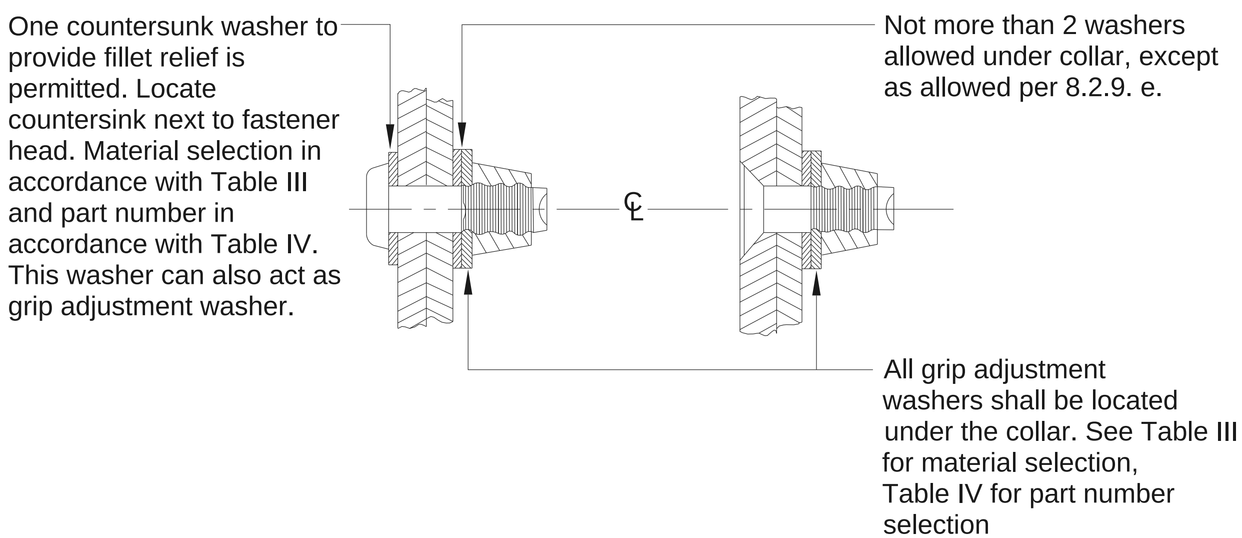


FIGURE 8 - SUMMARY OF WASHER USE LIMITATIONS FOR STEEL, ALUMINUM, AND TITANIUM STRUCTURES

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.4 USE OF WASHERS (Continued)

TABLE III - WASHER MATERIAL SELECTION FOR GRIP LENGTH ADJUSTMENT, FILLET RELIEF, AND COUNTERBORE SUBSTITUTION FL 2 FL 3 FL 4

COLLAR OR NUT MATERIAL	STRUCTURE MATERIAL	C'BORE. SUB. OR GRIP ADJUSTMENT WASHER (LOCATE NEXT TO NUT OR COLLAR)	FILLET RELIEF WASHER (LOCATE NEXT TO STRUCTURE)
		MATERIAL	MATERIAL
Aluminum	Aluminum	2024 Al Preferred. Cd. Plated CRES Optional	2024 or 7075
	Cd. Plated Steel		Cd. Plated CRES
	Cd. Plated CRES		CRES Unplated
	Titanium	CRES Unplated	CRES Unplated
	Magnesium	2024 Al.	5052 Al. FL 1
Cd. Plated Steel CRES Monel	Aluminum	2024 Al. or Cd. Plated CRES	2024 or 7075 Al. or Cd. plated CRES
	Steel		
	CRES	CRES Unplated	CRES Unplated
	Titanium	CRES Unplated	CRES Unplated
	Magnesium	2024 Al. or Cd. Plated CRES	5052 Al. FL 1
Unplated CRES Monel	Aluminum	Cd. Plated CRES	2024 or 7075 Al. or Cd. plated CRES
	Steel	Cd. Plated CRES	Cd. Plated CRES
	CRES	CRES Unplated	CRES Unplated
	Titanium	CRES Unplated	CRES Unplated
	Magnesium	Cd. Plated CRES	5052 Al. FL 1

FL 1 5052 aluminum washer is mandatory under protruding fastener head if head end bears on magnesium.

FL 2 See Section 8.2.4 for limitations.

FL 3 Cd is the abbreviation for Cadmium, Al. is the abbreviation for aluminum, and CRES is the abbreviation for corrosion resistant steel.

FL 4 See FL 3 of Table IV for use of aluminum washers with non-aluminum tension bolts or under nickel alloy 718 shear bolts.

TABLE IV - WASHER PART NUMBER REFERENCE TABLE FL 1 FL 2 FL 3

PLAIN WASHER MATERIAL	WASHER PART NUMBERS (REF)			
	NOMINAL	0.0156 O.S.	0.0312 O.S.	0.0468 O.S.
CRES Cd. PLATED	NAS1149E( )P BACW10BP( )DP BACW10BP( )NDP BACW10BN( )SP	BACW10BP( )1DP BACW10BP( )1NDP	BACW10BP( )2DP BACW10BP( )2NDP	USE NEXT LARGER STANDARD SIZE

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.4 USE OF WASHERS (Continued)

TABLE IV - WASHER PART NUMBER REFERENCE TABLE FL 1 FL 2 FL 3 (Continued)

PLAIN WASHER MATERIAL	WASHER PART NUMBERS (REF)			
	NOMINAL	0.0156 O.S.	0.0312 O.S.	0.0468 O.S.
CRES Cd. PLATED	BACW10P4( )CG FL 5	BACW10P4( )CG FL 5	USE NEXT LARGER STANDARD SIZE	USE NEXT LARGER STANDARD SIZE
2024 OR 7075 ALUMINUM	BACW10BN( )AP NAS1149D( )J BACW10AT3( )AS FL 4 BACW10P4( )AN FL 5	BACW10AW( )AS BACW10AW( )AST BACW10BN( )1AP BACW10AT3( )AS FL 4 BACW10P4( )AN FL 5	BACW10AW1( )AS BACW10AW1( )AST BACW10BN( )2AP - - - USE NEXT LARGER STANDARD SIZE	BACW10AW3( )AS BACW10AW3( )AST - - - USE NEXT LARGER STANDARD SIZE
5052 ALUMINUM	BACW10BN( )ANP NAS620A( ) NAS1197-( )	BACW10AW ( )AN BACW10AW( )ANT BACW10BN( )1ANP	BACW10AW1( )AN BACW10AW1( )ANT BACW10BN( )2ANP	BACW10AW3( )AN BACW10AW3( )ANT
CRES UNPLATED	BACW10BN( )UP BACW10BP( )APU BACW10BP( )PTU BACW10BP( )NAPU BACW10BP( )NPTU NAS620C( ) NAS1149E( )R	BACW10BP( )1APU BACW10BP( )1PTU BACW10BP( )1NAPU BACW10BP( )1NPTU	BACW10BP( )2APU BACW10BP( )2PTU BACW10BP( )2NAPU BACW10BP( )2NPTU	USE NEXT LARGER STANDARD SIZE.
CRES Cd PLATED	BACW10EG( )C BACW10CT( )C	BACW10EG( )1C BACW10CT( )1C	BACW10EG( )2C BACW10CT( )2C	USE NEXT LARGER STANDARD SIZE
2024 OR 7075 ALUMINUM	BACW10CT( )D BACW10CT( )J BACW10EG( )J	BACW10AW-C( )AS BACW10CT( )1J BACW10EG( )1J	BACW10AW-C1( )AS BACW10CT( )2J BACW10CT( )2D BACW10EG( )2J	BACW10AW-C3( )AS
5052 ALUMINUM	BACW10BN( )ANC	BACW10AW-C( )AN	BACW10AW-C1( )AN	BACW10AW-C3( )AN
CRES UNPLATED	BACW10CT( )CU BACW10EG( )CU	BACW10CT( )1CU BACW10EG( )1CU	BACW10CT( )2CU BACW10EG( )2CU	USE NEXT LARGER STANDARD SIZE

FL 1 ( ) indicates where diameter dash number is to be located in part number.

FL 2 All washer thicknesses listed in the corresponding specifications are acceptable. Please note, however, that all washers listed may not satisfy thickness requirements for use as counterbore substitution washers in accordance with Section 8.2.9.e.

FL 3 Aluminum washers shall not be used with:

- Non-aluminum tension hex drive bolts (for example, BACB30NX/BACB30NY) installed with tension collars (BACC30BH or BACC30X) or with nuts listed for tension bolts in Table XII

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.4 USE OF WASHERS (Continued)

- Non-aluminum tension lockbolts (for example, BACB30UA/BACB30UB) installed with tension collars (BACC30BF).
- Nickel alloy 718 shear or tension bolts (BACB30YK, BACB30YL, BACB30YM and BACB30YN) installed with BACC30CP or BACC30CQ collars or with BACN11AK nuts.

It is, however, acceptable to use aluminum washers with non-aluminum tension hex drive bolts (for example, BACB30NX, BACB30NY) with shear collars (for example, BACC30M or BACC30AB) in accordance with Engineering drawing, provided the washer material selection requirements of Table III are met.

FL 4 These washers are to be used under the concave base of self-aligning collars (BACC30AG, BACC30BQ) for grip adjustment.

FL 5 These washers are to be used under the concave base of self-aligning nuts (BACN10MT / BACW10AU) for counterbore substitution or for grip adjustment.

8.2.5 COLLARS

a. Collars as received are lubricated by the manufacturer. Degreasing or applying additional lubrication is not allowed.

b. Collars shall not be reused.

8.2.6 HI-SHEAR RIVET INSTALLATION

a. Hi-Shear rivets shall be driven with conventional riveting equipment. For aluminum collars, use the applicable ST1109 tool, or equivalent die.

b. Rivet sets shall not be tipped more than 5 degrees from the rivet axis during driving.

c. Collars shall be driven slightly beyond the point at which they trim off. See Section 11.3.1.b. and Section 11.3.1.c.

d. When an interference fit is specified, the following additional operations are required:

(1) Break edges of hole (0.0156 to 0.0312) on entering side for protruding head fasteners only.

(2) Drive rivet into the hole until the head is fully seated before swaging the collar.

e. Hi-Shear rivets located in faying seal areas shall have the protruding shank end cleaned of sealant after the fastener has been inserted through the structure and prior to installing the collar. Traces of sealant visible to the naked eye are acceptable.

f. The manufactured head shall not be hammered or peened to meet fastener head seating requirements.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.7 LOCKBOLT INSTALLATION - GENERAL

a. BACC30L, BACC30K, BACC30BE, BACC30BF, BACC30BK, NAS1080C04 and 05, NAS1080MG04 and 05, NAS1080K and NAS1080E04 and 05 require no special orientation. Other lockbolt collars shall be oriented as shown in Figure 9. Collars with self retaining features (for example, BACC30BK( )R and BACC30BF( )R( ) shall be oriented such that the retaining feature faces away from the structure. Do not use them inside fuel tank.

b. Pull-type shear and tension standard lockbolts shall be installed with pull tools fitted with swage dies having ST10-1027G, or equivalent, cavity configuration.

c. Pull-type and stump-type shear lightweight titanium lockbolts shall be installed with tools fitted with swage dies in accordance with Figure 10, or equivalent, cavity configuration.

d. Swaging dies shall not be tipped more than 3 degrees from lockbolt axis during installation.

e. Pull-type lockbolts with pin-tails removed are not valid substitute parts and shall not be used in place of stump type lockbolts.

f. Collar swaging and pin break-off shall be accomplished with a single stroke of the puller. No other method of pin break-off is permitted.

g. Sealant and/or foreign residue shall be removed from the annular grooves of the lockbolt prior to collar installation. Traces of sealant visible to the unaided eye are acceptable. BACC30BK, BACC30BE and BACC30BF aluminum collars shall be installed on the wet side of fuel tanks to provide fluid-tight installations, unless otherwise specified on the Engineering drawing. Sealant on the surface of the structure, visible to the unaided eyes, projecting from under the bearing surface of the collar, after swaging is not acceptable. This restriction does not apply when lockbolts with sealant escape grooves are used.

(1) If lockbolts with sealant grooves are used, the installation shall conform to the following requirements:

(a) These fasteners shall not be used in fluid-tight areas.

(b) Where the Engineering drawing calls out sealant escape groove lockbolts with wet sealant installation (e.g. BAC5000 Method 1, 2, or 3 installation), use BMS5-95, C-80 or C-168 sealant.

NOTE: Other classes of sealant may not flow properly through the sealant escape groove.

(c) Sealant on both the lock and pin-tail grooves shall be dry-wiped with a cloth before insertion of collar. It is not necessary, however, to thoroughly wipe the grooves.

(d) The swaging of collar shall be carried out within 4 hours of application of sealant.

h. All tension and shear (nontitanium) stump-type lockbolts shall be installed with swage dies having ST30-1110, or equivalent, cavity configuration.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.7 LOCKBOLT INSTALLATION - GENERAL (Continued)

i. Shear-head titanium stump-type lockbolts shall be installed by either squeeze or impact methods in accordance with Table V.

j. Shear-head titanium pull-type lockbolts shall be installed in accordance with Table VI.

k. Impact driving of NAS1080P collars is prohibited.

l. If lockbolt heads are to be overcoated with BMS5-62, BMS8-70, BMS8-78 or BMS8-103, then the bolts shall be vapor degreased in accordance with BAC5408, prior to installation.

m. Steel lockbolts may be installed on slopes up to 2 degrees (collar side) or up to 7 degrees when self-aligning washers are used. Titanium lockbolts may be installed on slopes up to 7 degrees (collar side).

n. When installing BACC30BK( )R and BACC30BF( )R lockbolt collars with the self retaining feature, traces of the retaining feature/adhesive visible to the unaided eye are acceptable on the collar and lockbolt pin upon completion of the fastener installation. The collar is to be installed with the locking tab facing away from the bolt head.

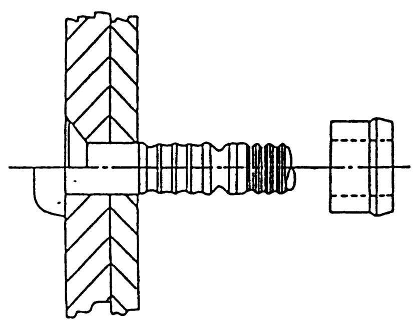


FIGURE 9 - COLLAR ORIENTATION

TABLE V - STUMP LOCKBOLT INSTALLATION TOOLS

LOCKBOLT PART NO.	COLLAR PART NO.	APPROVED INSTALLATION METHOD	
		PROCESS	DIE CAVITY CONFIGURATION
BACB30VP( )K( )	BACC30BK( )	Squeeze	ST10-1027J
BACB30VR()K( )	BACC30BK()	Squeeze	ST10-1027J
		Impact FL 2	- - -
BACB30XU()K( )	BACC30BK	Squeeze	ST10-1027J
BACB30UC()K( )	BACC30BE	Squeeze	ST10-1027G
BACB30UD()K( )	BACC30BE	Squeeze	ST10-1027G
BACB30UC6K( )	BACC30K6	Impact FL 1	ST10-1110B
		Squeeze	ST30-1110 or ST10-1110B
BACB30UD6K( )	BACC30K6	Impact FL 1	ST10-1110B
		Squeeze	ST30-1110 or ST10-1110B

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.7 LOCKBOLT INSTALLATION - GENERAL (Continued)

TABLE V - STUMP LOCKBOLT INSTALLATION TOOLS (Continued)

LOCKBOLT PART NO.	COLLAR PART NO.	APPROVED INSTALLATION METHOD	
		PROCESS	DIE CAVITY CONFIGURATION
BACB30UC8K( )	BACC30K8	Impact FL 1	ST10-1110B
		Squeeze	ST30-1110 or ST10-1110B
BACB30UD8K( )	BACC30K8	Impact FL 1	ST10-1110B
		Squeeze	ST30-1110 or ST10-1110B
BACB30UC10( )	NAS1080K10	Impact FL 1	ST10-1110B
		Squeeze	ST10-1110B
BACB30UD10K( )	NAS1080K10	Impact FL 1	ST10-1110B
		Squeeze	ST10-1110B

FL 1 Impact driving applies only to the Automatic Spar Assembly Tool (ASAT) and rework of stumps that have been installed by the ASAT.

FL 2 Impact driving applies only to Automated Spar Assembly Tool (ASAT). Reswaging prohibited.

TABLE VI - PULL LOCKBOLT INSTALLATION TOOLS

LOCKBOLT PART NO.	COLLAR PART NO.	APPROVED INSTALLATION METHOD	
		PROCESS	DIE CAVITY CONFIGURATION
BACB30TY( )K( )	BACC30BE( )	Pull Tools	ST10-1027G
BACB30TZ( )K( )	BACC30BE( )	Pull Tools	ST10-1027G
BACB30UA( )K( )	BACC30BF	Pull Tools	ST10-1027G
BACB30UB( )K( )	BACC30BF	Pull Tools	ST10-1027G
BACB30VM( )K( )	BACC30BK	Pull Tools	ST10-1027H
BACB30VN( )K( )	BACC30BK	Pull Tools	ST10-1027H
BACB30XT( )K( )	BACC30BK	Pull Tools	ST10-1027H

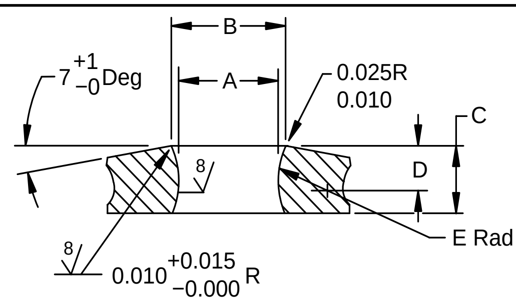


FIGURE 10 - LIGHTWEIGHT COLLAR SWAGE DIE CONFIGURATION (INCH)

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.7 LOCKBOLT INSTALLATION - GENERAL (Continued)

NOMINAL DIA.	A		B	C	D	E
	+0.0010 -0.0010 (NEW DIES) FL 1	MAX. IN-SERVICE DIA. FL 2	+0.0030 -0.0000	+0.015 -0.005	REF	(RADIUS) ± 0.005
5/32	0.2219	0.2239	0.2580	0.103	0.0661	0.126
3/16	0.2450	0.2470	0.2813	0.120	0.0670	0.130
1/4	0.3241	0.3261	0.3722	0.157	0.0928	0.188
5/16	0.4062	0.4082	0.4667	0.195	0.1201	0.250
3/8	0.4797	0.4827	0.5525	0.233	0.1469	0.311

FIGURE 10 - LIGHTWEIGHT COLLAR SWAGE DIE CONFIGURATION (INCH) (Continued)

FL 1 New swage dies must conform to the dimensions shown in this column.

FL 2 Used swage dies shall not exceed the dimensions shown in this column.

8.2.8 LOCKBOLT INSTALLATION FOR FASTENERS INSTALLED BY PULL-IN PROCESS

Pull-in process may be used for seating pull type lockbolts (BACB30VM, BACB30VN and BACB30XT only). The process involves gripping the pin-tail and pulling the lockbolt into place (as against driving the head of the fastener with a rivet hammer). The process requires use of self-release nose-pieces and pressure regulators, to control the force applied to the pin-tail. After the fastener is seated, collar is put on the pin-tail and swaged using the regular swage tools. If this process is used, following are mandatory:

a. The slope of the structure on the pin-tail side shall be less than 2 degrees.

b. Head to shank fillet relief in accordance with Table II is required in all fastener holes regardless of material and size. Fillet relief washers may be used in lieu of fillet relief, for protruding head fasteners.

c. The countersink and hole intersection should blend smoothly at maximum hole size (see Figure 3). The maximum countersink to hole angularity and eccentricity for flush head fasteners shall be in accordance with Figure 11. To ensure these requirements are met, and to ensure proper seating, use of an integral drill/countersink tool is mandatory. To meet flushness requirements, for lockbolts installed by pull-in process, the countersink depth may be increased by an additional 0.010 inch maximum using a piloted and radiused tool.

d. Joint shall be firmly clamped so that there are no gaps at the faying surfaces at the time of bolt installation (clamping force in accordance with BAC5300).

e. Fastener shall be driven into the structure with a soft face mallet until the pintail protrudes beyond the structure such that the nose piece of the puller can properly engage. Remove sealant from pull grooves. During bolt pull-in, the nose piece of the puller shall not slip on the pull grooves. There shall be no visible damage to pull grooves or the structure after the bolt is seated.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.8 LOCKBOLT INSTALLATION FOR FASTENERS INSTALLED BY PULL-IN PROCESS (Continued)

f. Fastener head shall be seated in the structure prior to collar installation. For protruding head bolts, the gap requirement of Section 11.2.1 apply. No measurable gap is allowed between bearing surface of the bolt head of flush head lockbolt and countersink. Refer to Figure 12. However, tolerances in the countersinking operation and in the heads of flush head fasteners may result in partially exposed countersinks. Exposed areas around flush fastener heads are acceptable if the hole tolerances, fastener tolerances, head gapping, and flushness requirements are met. Refer to Figure 4. Impact drive unseated and not fully seated fasteners.

g. Remove sealant from lockgrooves before the collar is installed.

h. Collar swaging and pin break-off shall be accomplished with a single stroke of the puller. No other method of pin break-off is permitted.

i. Pull-in lockbolt nose piece shall be selected in accordance with Table VII.

j. The hydraulic power unit shall be adjusted using a Skidmore-Wilhelm clamp pressure gage or similar device, to provide pull-in forces listed in Table VIII.

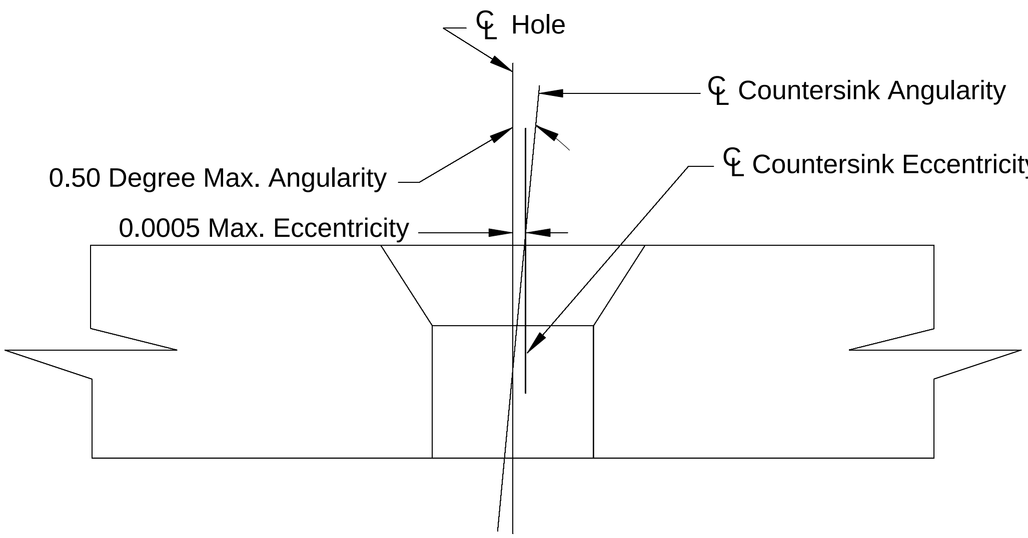


FIGURE 11 - COUNTERSINK ANGULARITY AND ECCENTRICITY

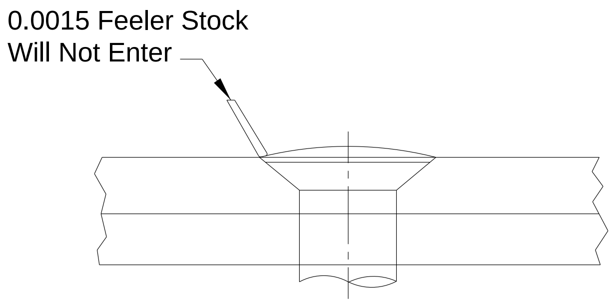


FIGURE 12 - COUNTERSINK FIT

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.8 LOCKBOLT INSTALLATION FOR FASTENERS INSTALLED BY PULL-IN PROCESS (Continued)

TABLE VII - PULL-IN NOSE ASSEMBLY
FL 1

FASTENER DIAMETER	NOSE ASSEMBLY
3/16	99-1826 99-1826-1 99-1829 99-1829-1 99-1834 99-1834-1
1/4	99-1827 99-1827-1 99-1830 99-1830-1 99-1835 99-1835-1
5/16	99-1813 99-1831 99-1831-1 99-1836
3/8	99-1832 99-1832-1 99-1837

FL 1 99-18XX nose assemblies are available from ALCOA Fastening Systems - Commercial Products, Kingston, New York

TABLE VIII - PULL-IN FORCE

FASTENER DIAMETER	MAXIMUM PULL-IN FORCE, POUNDS
3/16	1450
1/4	2250
5/16	3550
3/8	5100

8.2.9 HEX-DRIVE BOLT INSTALLATION

a. All hex drive bolts installed in transition or interference fit holes shall have the manufactured head seated to the structure, prior to collar installation.

CAUTION If there is a gap in the joint during fastener insertion and the pin protrusion is at or near the maximum allowed in Section 11.3.3.a. prior to collar installation, then the pin protrusion may be rejectable following installation.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9 HEX-DRIVE BOLT INSTALLATION (Continued)

b. Collars shall be turned onto the hex drive bolt until the turning action causes the collar hex to shear off. No other method of hex removal is permitted, and no further wrenching of the collar is permitted. The sheared off hex is to be discarded.

c. Light weight collars shall not be used on standard weight hex drive bolts nor shall standard weight collars be used on light weight hex drive bolts.

d. Lockbolt collars shall not be used on hex drive bolts, except as provided in Section 8.2.10.

e. When the Engineering drawing requires or allows the use of non-counterbored self-locking nuts in lieu of mating collars on hex-drive bolts, washers shall be used for counterbore substitution. See Section 8.2.4 for washer selection. Examples of non-counterbore nuts: BACN10GW, BACN10JC, BACN10YR, BACN11Z, NASM21042, NASM21043, NAS679, NAS1291, NAS1804 and NAS1805.

(1) For lightweight (BACB30VT, BACB30VU and BACB30YP) hex drive bolts, one 0.063 inch and one 0.032 inch thick washers shall be used. Place 0.063 inch thick washer against structure. Ensure that the full chamfer of the bolt protrudes out of the nut. Exception: In cases where the chamfer does not fully protrude out, use one 0.063 inch washer only. For 5/32 inch diameter bolts only, it is permissible to use two 0.032 inch thick washers in lieu of one 0.063 inch washer.

(2) For standard hex-drive bolts, washers shall be used in accordance with Table IX. (Examples of standard hex-drive bolts: BACB30FM, BAC30FN, BACB30JC, BACB30MY, BACB30MB, BACB30ND, BACB30NW, BACB30NX, BACB30NY, BACB30YK, BACB30YL, BACB30YM, BACB30YN, HLT 420, HLT 421, HLT 422, HLT 423).

(3) Counterbore substitution washers are in addition to grip adjustment and fillet relief washers (see Figure 13).

(4) Counterbore substitution washers are required when non-counterbored nuts are used on top of the self-aligning washers (for example, BACW10CA). (See Section 11.3.3.c. for pin protrusion).

(5) 0.063 thick counterbore substitution washers (BACW10P416CG through BACW10P421CG (cadmium plated CRES) or BACW10P416AN through BACW10P421AN (anodized aluminum)) shall be used under the concave base of the self-aligning BACN10MT/BACW10AU nut/washer combination. (See Figure 6).

(6) If conflict exists between washers specified on the Engineering drawing and the washers required by this specification for counterbore substitution, contact Engineering Liaison.

(7) Ensure that the nuts do not bottom out on the threads.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9 HEX-DRIVE BOLT INSTALLATION (Continued)

TABLE IX - COUNTERBORE SUBSTITUTION WASHERS FOR STANDARD WEIGHT HEX-DRIVE FASTENERS FL 1

NOMINAL SIZE OF BOLT DIAMETER INCH	NUMBER AND THICKNESS OF WASHER-NOMINAL, 0.0156 AND 0.0312 OVERSIZE	NUMBER AND THICKNESS OF WASHER -0.0468 OVERSIZE
5/32 through 3/8 (-5 through -12)	Two 0.063 inch (Nominal) FL 2	Three 0.063 inch (Nominal) FL 7
7/16 through 9/16 (-14 through -18)	Two 0.071 inch (Nominal) FL 3	Three 0.071 inch (Nominal) FL 8
5/8 (-20)	Two 0.080 inch (Nominal) FL 4	Two 0.080 inch (Nominal) FL 4
3/4 (-24)	Three 0.080 inch (Nominal) FL 5	Three 0.080 inch (Nominal) FL 5
7/8 and 1 (-28 through -32)	Three 0.090 inch (Nominal) FL 6	Three 0.090 inch (Nominal) FL 6

FL 1 Place the thicker washer against the structure.

FL 2 The chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use one 0.063 thick washer and one 0.032 thick washer. If even then the chamfer does not protrude out, use one 0.063 thick washer. For 5/32 inch diameter bolts only, it is permissible to use two 0.032 inch thick washers in lieu of one 0.063 inch thick washer.

FL 3 The chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use one 0.071 thick washer and one 0.032 thick washer. If even then the chamfer does not protrude out, use one 0.071 thick washer.

FL 4 The chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use one 0.080 thick washer and one 0.032 thick washer. If even then the chamfer does not protrude out, use one 0.080 thick washer.

FL 5 The chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use two 0.080 thick washers.

FL 6 Chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use two 0.090 thick washers.

FL 7 The chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use two 0.063 thick washer and one 0.032 thick washer. If even then the chamfer does not protrude out, use two 0.063 thick washer.

FL 8 The chamfer of the bolt shall always protrude out beyond the nut. In cases where the chamfer does not protrude out, use two 0.071 thick washer and one 0.032 thick washer. If even then the chamfer does not protrude out, use two 0.071 thick washer.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9 HEX-DRIVE BOLT INSTALLATION (Continued)

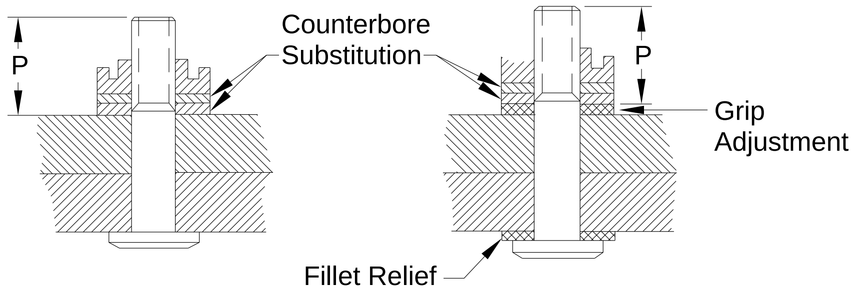


FIGURE 13 - USE OF WASHERS WITH NON-COUNTERBORED NUTS

NOTE: P dimension is pin protrusion.

f. Washers may not be used between self-sealing collars/nuts and structure.

g. Self-sealing BACN10WM, BACN10TN, and BACN11BT nuts and self-sealing BACC30BP collars shall not be installed against slopes greater than 2 degrees.

h. Self-sealing collars or nuts must be installed on the fluid side to provide a fluid-tight installation unless otherwise specified on the Engineering drawing. Sealant and/or foreign residue must be removed from the threads of the bolt prior to seal nut or seal collar installation. Permitted installation conditions include the following:

(1) Traces of sealant visible to the unaided eye.

(2) Traces of sealant on the surface of the structure, which do not form a continuous ring, projecting from under the bearing surface of the seal nut or seal collar.

i. Teflon must not extrude beyond the outer periphery of the base of the seal nut or seal collar after installation.

j. Self-sealing collars and self-sealing nuts shall not be reused.

k. When installing a hex drive bolt use a hex key to prevent bolt rotation while installing the collar or nut.

l. Collars and nuts shall not be reused.

m. Hex-drive bolts shall be replaced with new hex-drive bolts when removed collars or nuts are Alloy Steel, CRES or Titanium.

n. If removing aluminum collars or nuts from completed assemblies or installations, hex-drive bolt shall be visually examined for thread damage prior to installing new collars or nuts. If any thread damage exists, the hex-drive bolt shall be replaced with a new hex-drive bolt.

o. Where the Engineering drawing requires that torque be re-applied, use the following procedure:

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9 HEX-DRIVE BOLT INSTALLATION (Continued)

(1) The original torque application for permanent or temporary fasteners shall be completed with at least 60 minutes (20 minutes for B-1/2 Class Sealants) of squeeze-out life remaining in fay surface and fastener sealant. Temporary fastener usage shall be in accordance with the applicable fay surface sealant process specification. Re-apply torque in accordance with the applicable torque table using a torque tool setpoint from the appropriate column for re-torque application.

(2) Commence re-application of torque not sooner than 10 minutes for Class C-20, C-24, or C-48 sealant, and not sooner than 60 minutes for sealants with squeeze-out life longer than 48 hours. Complete re-application of torque for permanent or temporary fasteners before the squeeze-out life of fay surface and fastener sealant expires. Temporary fastener usage shall be in accordance with the applicable fay surface sealant process specification.

NOTE: Many fay sealed joints do not require torque re-application. For those installations, apply torque using the appropriate torque tool setpoint from Table X columns for fay sealed joint with no re-torque requirement. Previous revisions of this specification had instructions to re-apply torque, when required by the Engineering drawing, using the upper end of the torque range for the second torque application. When the torque tool setpoints replaced the torque range, the re-torque requirement was placed in the Table X columns headed “FAY SEALED WITH RETORQUE REQUIRED”.

8.2.9.1 TORQUE REQUIREMENTS

a. All tools and procedures used to apply torque must conform to the requirements given in BSS7083 unless otherwise specified on the Engineering drawing

b. Tighten all locknuts installed on hex-drive bolts (nominal and oversize) using the torque tool setpoint value specified on the applicable torque table. The indicated value must be within the specified tolerance for a given torque tool setpoint.

c. A ±3 percent tolerance is allowed for the torque tool setpoint in the torque tables. Use this procedure to calculate the tolerance.

(1) Multiply the setpoint value by .03.

(2) Add and subtract this value to find the maximum and minimum torque values for the tolerance range.

(3) Round up to the nearest whole number if the value is .5 or more. Round down to the nearest whole number if the value is less than .5. For example: If the setpoint value = 60 inch-lb.

60 x .03=1.8
60 - 1.8=58.2, round down to 58
60 + 1.8=61.8, round up to 62
The ±3 percent tolerance is 58 -62 inch-lbs

d. When the bolt part number and the nut part number are not on the same torque table, apply the torque from the table with the lowest setpoint value.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9.1 TORQUE REQUIREMENTS (Continued)

e. When torque tool adapters, extensions, or universal joints are used, torque indicator reading correction factors may be required. Refer to BSS7083 for adapter/extension usage and the procedures used to obtain the correction factors.

TABLE X - HEX DRIVE TORQUE: LIGHTWEIGHT SHEAR BOLTS FL 4

FASTENER CLASS	LIGHTWEIGHT SHEAR HEX-DRIVE BOLTS			
FASTENER PART NUMBER	BACB30VT, BACB30VU, BACB30YP			
NUT PART NUMBER	BACN10YZ FL 1, BACN10ZV FL 1 BACN11BT FL 1 FL 3		BACN11AG, BACN10JC, BACN10YR, BACN11AJ FL 1, MS21042, MS21043, NAS679, NAS1291	
NOMINAL FASTENER DIAMETER (INCH)	TORQUE TOOL SETPOINT (INCH-LB)		TORQUE TOOL SETPOINT (INCH-LB)	
	NO FAY SEAL OR FAYSEALED JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL2	NO FAY SEAL OR FAYSEALED JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL2
0.1640	20	25	20	25
0.1900	30	35	30	35
0.2500	70	80	70	80
0.3125	125	140	145	160
0.3750	180	200	220	240
0.4375	-	-	300	330

FL 1 Counterbored nuts. Counterbore substitution washers are not required.

FL 2 Use the torque value in this column when the Engineering drawing requires a retorque. Apply the applicable setpoint value for the initial torque and re-torque unless otherwise specified on the Engineering drawing.

FL 3 Use of washer(s) between self sealing nuts and structures is not permitted unless otherwise specified on the Engineering drawing.

FL 4 The tolerance on the torque tool setpoint is ±3%. Refer to Section 8.2.9.1.c. to calculate the min/max values.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9.1 TORQUE REQUIREMENTS (Continued)

TABLE XI - HEX-DRIVE TORQUE: STANDARD WEIGHT SHEAR BOLTS FL 4

FASTENER CLASS	STANDARD WEIGHT SHEAR HEX DRIVE BOLTS							
FASTENER PART NUMBER	BACB30FM, BACB30FN, BACB30MY, BACB30ND, BACB30NW, BACB30NZ, HL360, HL834				BACB30FM, BACB30NW, BACB30FN, BACB30NZ, BACB30MY, BACB30ND, HL360, HL834		BACB30YL, BACB30YN, HLT420, HLT421	
NUT PART NUMBER	BACN11AM FL 1, BACN10GW, BACN10JC, BACN10MT, BACN10TN FL 1, FL 3, BACN10YR, BACN10XJ FL 1, BACN10YT FL 1, BACN11BH FL 1, BACN11E FL 1, BACN11Z, H600, KFN305 FL 1, KFN511 FL 1, KFN600, MS21042, MS21043, NAS679, NAS1291, NAS1804, NAS1805		BACN10WM		BACN10ZZ FL 1 KFN609 FL 1		BACN11AK FL 1, BACN11Z, NAS1804, NAS1805	
NOMINAL FASTENER DIAMETER (INCH)	TORQUE TOOL SETPOINT (INCH-LBS)		TORQUE TOOL SETPOINT (INCH-LBS)		TORQUE TOOL SETPOINT (INCH-LBS)		TORQUE TOOL SETPOINT (INCH-LBS)	
	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2
0.1640	20	25	-	-	20	25	-	-
0.1900	30	35	-	-	25	30	35	40
0.2500	70	80	70	80	70	75	75	85
0.3125	145	160	135	145	135	150	160	175
0.3750	220	240	205	210	195	210	220	240
0.4375	300	330	-	-	300	330	315	340
0.5000	400	430	-	-	400	430	470	505
0.5625	540	575	-	-	-	-	-	-
0.6250	665	700	-	-	-	-	-	-
0.7500	950	1000	-	-	-	-	-	-

FL 1 Counterbored nuts. Counterbore substitution washers are not required.

FL 2 Use the torque value in this column when the Engineering drawing requires a re-torque. Apply the applicable setpoint value for the initial torque and re-torque unless otherwise specified on the Engineering drawing.

FL 3 Use of washer(s) between self sealing nuts and structures is not permitted unless otherwise specified on the Engineering drawing.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.9.1 TORQUE REQUIREMENTS (Continued)

FL 4 The tolerance on the torque tool setpoint is ±3%. Refer to Section 8.2.9.1.c. to calculate the min/max values.

TABLE XII - HEX-DRIVE TORQUE: STANDARD WEIGHT TENSION BOLTS FL 3

FASTENER CLASS	STANDARD WEIGHT TENSION HEX-DRIVE BOLTS					
FASTENER PART NUMBER	BACB30JC, BACB30MB, BACB30NX, BACB30NY		BACB30YK, BACB30YM, HLT422, HLT423		BACB30YK, BACB30YM, HLT422, HLT423	
NUT PART NUMBER	BACN11AM FL 1, BACN11BH FL 1, BACN11E FL 1, BACN10GW, BACN10MT, BACN10YT FL 1, BACN11Z, KFN305 FL 1, KFN511 FL 1, MS21042, NAS1804, NAS1805		BACN11Z, NAS1804, NAS1805		BACN11AK FL 1	
NOMINAL FASTENER DIAMETER (INCH)	TORQUE TOOL SETPOINT (INCH-LB)		TORQUE TOOL SETPOINT (INCH-LB)		TORQUE TOOL SETPOINT (INCH-LB)	
	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2	NO FAY SEAL OR FAY SEAL JOINT AND NO RETORQUE	FAY SEALED JOINT AND RETORQUE FL 2
0.1640	30	35	30	35	-	-
0.1900	35	40	35	40	45	50
0.2500	90	95	90	95	120	130
0.3125	175	200	175	200	225	250
0.3750	310	360	310	360	390	420
0.4375	435	480	435	480	600	675
0.5000	720	800	720	800	865	1000
0.5625	820	900	820	900	1125	1250
0.6250	900	1000	900	1000	1450	1550
0.7500	1480	1650	1480	1650	-	-

FL 1 Counterbored nuts. Counterbore substitution washers are not required.

FL 2 Use the torque value in this column when the Engineering drawing requires a re-torque. Apply the applicable setpoint value for the initial torque and re-torque unless otherwise specified on the Engineering drawing.

FL 3 The tolerance on the torque tool setpoint is ±3%. Refer to Section 8.2.9.1.c. to calculate the min/max values.

8.2.10 HEX-DRIVE BOLT INSTALLATION USING MACHINE SWAGED LOCKBOLT COLLARS

This is a process where lockbolt collars are swaged on hex drive bolts. This process is limited to the types and sizes shown below when the process outlined is followed:

a. This process is limited to the automated installation of 3/16, 1/4 and 5/16 diameter BACB30YP or BACB30VT shear, lightweight hex-drive fasteners with BACC30BK lightweight lockbolt collars by automatic swaging equipment certified in accordance with D6-56617 as allowed by Engineering drawing. Manual installation is not allowed. Use of electromagnetic swaging is not allowed.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.10 HEX-DRIVE BOLT INSTALLATION USING MACHINE SWAGED LOCKBOLT COLLARS (Continued)

b. Collars with the ”R” code retaining feature shall not be used.

c. Hex-drive bolts with machine swaged collars are not fluid tight and are not acceptable for use in wing box applications or other fuel containment structure.

d. Swage dies used for this process shall comply with Figure 11.

e. Sealant on the threads prior to installation is acceptable, provided the collar is installed before the sealant squeeze out life has expired. Sealant extrusion from under the collar or around the exposed threads after installation is acceptable.

f. Collars may be installed on slopes up to 7 degrees (collar side).

g. Hex-drive bolts and collars shall not be reused after removal.

h. Slivers of collar material visible above the bolt point are acceptable.

NOTE: Slivers appear to be caused by lower tool misalignment and may indicate a need for machine maintenance.

i. Hole sizes shall be targeted for the lower third of the hole diameter tolerance, such that a positive interference fit is achieved.

j. Lightweight Hex-Drive Bolts with swaged collars shall be installed in accordance with Table XXIX and Figure 32. Swage height and pin protrusion shall be verified using the applicable GO-NO-GO gage as shown in Figure 32.

8.2.11 FASTENER REPLACEMENT-OVERSIZE

Hi-shear rivets may be replaced with larger diameter fasteners in accordance with the limitations shown in Table XIII. Lockbolts and hex drive fasteners may be replaced with oversize fasteners in accordance with Table XIV. The hole sizes and edge margin requirements for transition fit, close ream and Class I holes shall be in accordance with Table XV . For interference fit installations, they shall be in accordance with Table XVI.

CAUTION When oversizing for the flush head fasteners, consider the extra depth. If the sheet thickness is too small, there may be knife edge condition, which is unacceptable.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XIII - HOLE SIZES FOR OVERSIZE HI-SHEAR RIVETS

ORIGINAL FASTENER	**OVERSIZE FASTENER**	**NOMINAL OVERSIZE AND NOTES**	**ORIGINAL FASTENER NOMINAL DIAMETER**							
			(-6) No. 10		**(-8) 0.250**		**(-10) 0.313**		**(-12) 0.375**	
			ORIGINAL HOLE SIZE							
			0.190 0.187 or 0.189 0.185	**0.192 0.189**	**0.250 0.247 or 0.249 0.245**	**0.254 0.250**	**0.313 0.309 or 0.312 0.308**	**0.317 0.313**	**0.375 0.371 or 0.374 0.370**	**0.379 0.375**
			HOLE SIZES FOR OVERSIZE REPLACEMENT FASTENERS							
BACR15AY BACR15AZ	BACR15BE BACR15BG	0.0156 FL 1	0.203 0.200	0.207 0.203	0.266 0.263	0.270 0.266	0.328 0.325	0.332 0.328	0.391 0.388	0.395 0.391
	BACB30NE-X BACB30LU-X	0.0156 FL 1	0.2036 0.2026		0.2661 0.2651		0.3286 0.3276		0.3911 0.3901	
	BACR15BF BACR15BH	0.0312 FL 2	0.219 0.216	0.223 0.219	0.281 0.278	0.285 0.281	0.344 0.341	0.348 0.344	0.406 0.403	0.410 0.406
	BACB30NE-Y BACB30LU-Y	0.0312 FL 2	0.2192 0.2182		0.2817 0.2807		0.3442 0.3432		0.4067 0.4057	
BACR15AU	BACR15AX	0.0156 FL 1	0.203 0.200	0.207 0.203	0.266 0.263	0.270 0.266	0.328 0.325	0.332 0.328	0.391 0.388	0.395 0.391
		0.0312 FL 2	0.219 0.216	0.223 0.219	0.281 0.278	0.285 0.281	0.344 0.341	0.348 0.344	0.406 0.403	0.410 0.406

ORIGINAL FASTENER	**OVERSIZE FASTENER**	**NOMINAL OVERSIZE AND NOTES**	**ORIGINAL FASTENER NOMINAL DIAMETER**							
			(-14) 0.438		**(-16) 0.500**		**(-18) 0.563**		**(-20) 0.625**	
			ORIGINAL HOLE SIZE							
			0.438 0.434 or 0.437 0.433	**0.442 0.438**	**0.500 0.496 or 0.499 0.495**	**0.504 0.500**	**0.563 0.559 or 0.562 0.558**	**0.568 0.563**	**0.625 0.621 or 0.624 0.620**	**0.630 0.625**
			HOLE SIZES FOR OVERSIZE REPLACEMENT FASTENERS							
BACR15AY BACR15AZ	BACR15BE BACR15BG	0.0156 FL 1	0.453 0.450	0.458 0.453	0.516 0.513	0.521 0.516	0.578 0.575	0.583 0.578	0.640 0.637	0.646 0.641
	BACB30NE-X BACB30LU-X	0.0156 FL 1	0.4536 0.4526		0.5161 0.5151		0.5786 0.5776		0.6411 0.6401	
	BACR15BF BACR15BH	0.0312 FL 2	0.469 0.466	0.474 0.469	0.531 0.528	0.536 0.531	0.593 0.590	0.598 0.593	0.656 0.653	0.661 0.656
	BACB30NE-Y BACB30LU-Y	0.0312 FL 2	0.4692 0.4682		0.5317 0.5307		0.5942 0.5932		0.6567 0.6557	

FL 1 Quality Assurance verification required

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

FL 2 Engineering approval and Quality Assurance verification are required if edge margin is less than two diameters, whereas only Quality Assurance verification is required if the edge margin is twice the diameter of the original fastener, or greater.

TABLE XIV - ORIGINAL BOLT TO OVERSIZE BOLT CROSS REFERENCE FOR LOCKBOLTS AND HEX DRIVE FASTENERS FL 12 FL 13

ORIGINAL FASTENER	OVERSIZE FASTENER FL 1		
	0.0156 OVERSIZE FL 2 FL 6	0.0312 OVERSIZE FL 2 FL 7	0.0468 OVERSIZE FL 2 FL 8
BACB30AR	BACB30CU	- - -	- - -
	BACB30FP	BACB30KK	BACB30MA
	BACB30GW( )X	- - -	- - -
BACB30AT	BACB30CT	- - -	- - -
	BACB30GY( )X		
	BACB30FQ	BACB30KF FL 4	BACB30LW FL 4
BACB30DX	BACB30DX( )X	BACB30MD	- - -
	BACB30MC	BACB30MD	BACB30ME
BACB30DY	BACB30DY( )X		
	BACB30ND( )X	BACB30ND( )Y	- - -
BACB30FM6	BACB30MC6 FL 11	BACB30MD6 FL 11	BACB30MA6
	BACB30FP6 FL 3	BACB30KK FL 3	
BACB30FM FL 10	BACB30FP	BACB30KK	BACB3MA
BACB30FN	BACB30FQ	BACB30KF FL 4	BACB30LW FL 4
BACB30GP	BACB30GZ( )X FL 9	BACB30GZ( )Y FL 9	BACB30GZ( )Z FL 9
BACB30GQ	NONE	- - -	- - -
BACB30GW	BACB30CU	- - -	- - -
	BACB30GW( )X	- - -	- - -
	BACB30FP	BACB30KK	BACB30MA
BACB30GX	BACB30CH	- - -	- - -
	BACB30HX	- - -	- - -
	BACB30LX	BACB30LY	BACB30LZ
BACB30GY	BACB30GY( )X	- - -	- - -
	BACB30CT	- - -	- - -
	BACB30FQ	BACB30KF FL 4	BACB30LW FL 4
BACB30GR	BACB30GZ( )X	BACB30GZ( )Y	BACB30GZ( )Z
BACB30GS	None	- - -	- - -
BACB30GZ	BACB30GZ( )X	BACB30GZ( )Y	BACB30GZ( )Z
BACB30HA	None	- - -	- - -
BACB30HC	BACB30DX( )X	- - -	- - -
	BACB30MC	BACB30MD	BACB30ME
BACB30HD	BACB30DY( )X	- - -	- - -
	BACB30ND( )X	BACB30ND( )Y	- - -

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XIV - ORIGINAL BOLT TO OVERSIZE BOLT CROSS REFERENCE FOR LOCKBOLTS AND HEX DRIVE FASTENERS FL 12 FL 13 (Continued)

ORIGINAL FASTENER	OVERSIZE FASTENER FL 1		
	0.0156 OVERSIZE FL 2 FL 6	0.0312 OVERSIZE FL 2 FL 7	0.0468 OVERSIZE FL 2 FL 8
BACB30HG	BACB30CH	- - -	- - -
	BACB30LX	BACB30LY	BACB30LZ
BACB30JC	BACB30LX	BACB30LY	BACB30LZ
BACB30LD	None	- - -	- - -
BACB30MB	BACB30MC	BACB30MD	BACB30ME
BACB30MM	BACB30NW( )X	BACB30NY( )Y FL 4	- - -
BACB30MY6	BACB30NX 6X FL 11	BACB30NX6Y FL 11	- - -
BACB30MY FL 10	BACB30MY( )X	BACB30MY( )Y	- - -
BACB30ND	BACB30ND( )X	BACB30ND( )Y FL 4	- - -
BACB30NW	BACB30NW( )X	BACB30NW( )Y	- - -
BACB30NX	BACB30NX( )X	BACB30NX( )Y	- - -
BACB30NY	BACB30NY( )X	BACB30NY( )Y	- - -
BACB30NZ	BACB30NZ( )X	BACB30NZ( )Y	- - -
BACB30TY	BACB30NW( )X	BACB30NW( )Y FL 4	- - -
BACB30TZ	BACB30MY( )X	BACB30MY( )Y	- - -
BACB30UA	BACB30NY( )X	BACB30NY( )Y	- - -
BACB30UB	BACB30NX( )X	BACB30NX( )Y	- - -
BACB30UC	BACB30NW( )X	BACB30NW( )Y	- - -
BACB30UD6	BACB30NX 6X FL 11	BACB30NX 6Y FL 11	- - -
BACB30UD FL 10	BACB30MY( )X	BACB30MY( )Y	- - -
BACB30VM	BACB30NW( )X FL 5	BACB30NW( )Y FL 4	- - -
BACB30VN6	BACB30NX 6X FL 11	BACB30NX 6Y FL 11	- - -
BACB30VN FL 10	BACB30MY( )X	BACB30MY( )Y	- - -
BACB30VP	BACB30NW( )X FL 5	BACB30NW( )Y FL 4	- - -
BACB30VR6	BACB30NX 6X FL 11	BACB30NX 6Y FL 11	- - -
BACB30VR FL 10	BACB30MY( )X	BACB30MY( )Y	- - -
BACB30VT6	BACB30NX6X FL 11	BACB30NX6Y FL 11	- - -
BACB30VT FL 10	BACB30MY( )X	BACB30MY( )Y	- - -
BACB30VU	BACB30NW( )X	BACB30NW( )Y FL 4	- - -
BACB30XT	BACB30NW( )X	BACB30NW( )Y	
BACB30YP	BACB30NW( )X FL 5	BACB30NW( )Y	- - -
BACB30YK	BACB30YK( )X	BACB30YK( )Y	- - -
BACB30YL6	BACB30YK6X	BACB30YK6Y	- - -
BACB30YL FL 10	BACB30YL( )X	BACB30YL( )Y	- - -
BACB30YM	BACB30YM( )X	BACB30YM( )Y	- - -
BACB30YN	BACB30YN( )X	BACB30YN( )Y	- - -

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XIV - ORIGINAL BOLT TO OVERSIZE BOLT CROSS REFERENCE FOR LOCKBOLTS AND HEX DRIVE FASTENERS FL 12 FL 13 (Continued)

ORIGINAL FASTENER	OVERSIZE FASTENER FL 1		
	0.0156 OVERSIZE FL 2 FL 6	0.0312 OVERSIZE FL 2 FL 7	0.0468 OVERSIZE FL 2 FL 8
NAS1466-1472	BACB30DX( )X	- - -	- - -
NAS1476-1482	BACB30DY( )X	- - -	- - -
NAS1496-1502	BACB30DX( )X	- - -	- - -
HL360	HL560	HL960	- - -
HL834	HL924	HL942	- - -
HLT420	HLT620	HLT714	- - -
HLT421	HLT621	HLT716	- - -
HLT422	HLT622	HLT716	- - -
HLT423	HLT623	HLT717	- - -

FL 1 Fastener head style, alloy, and finish shall be identical to drawing requirement for nominal size.

FL 2 For edge margin requirements, see Table XVI and Table XVII.

FL 3 Though it is acceptable to use this shear fastener for this size, tension fastener is the preferred option. See FL 11

FL 4 Indicates those oversize fasteners which require deeper countersinks than the original fastener.

FL 5 Due to variations in head dimensions countersink depth adjustment may be required when installing oversize fasteners.

FL 6 For 0.0156 oversize collar, see Table XV

FL 7 For 0.0312 oversize collar, see Table XV

FL 8 For 0.0468 oversize collar, see Table XV

FL 9 For -6 and -8 sizes only

FL 10 All sizes except -6

FL 11 Use of tension fastener with shear collar or nut is the preferred option for this size.

FL 12 There are no .0156 oversize for -5 (0.164) size fasteners. For .0312 oversize use -6. For .0468 oversize use the 0.156 oversize for -6. See FL 11.

FL 13 For oversize replacement of lockbolts with hiloks, contact Liaison Engineering to determine proper collar selection.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XV - ORIGINAL COLLAR OR NUT TO OVERSIZE COLLAR OR NUT CROSS REFERENCE FOR HEX DRIVE FASTENERS

ORIGINAL COLLAR	OVERSIZE COLLARS/NUTS		
	0.0156	0.0312	0.0468
BACC30AB( )C	BACC30AB( )CY	BACC30AB( )CY	- - -
BACC30AB( )P	BACC30AB( )PY	BACC30AB( )PY	- - -
BACC30AB( )S	BACC30AB( )SY	BACC30AB( )SY	- - -
BACC30AG( )	BACC30AG( )	- - -	- - -
BACC30BP( )	BACC30M( ) FL 2	BACC30R( ) FL 2	BACC30Y( ) FL 2
BACC30BH( )	BACC30BH( )	BACC30AC( )	BACC30U( )
BACC30BH( )( ) FL 1	BACC30BH( )( ) FL 1	- - -	- - -
BACC30BL( )	BACC30M( )	BACC30R( )	BACC30Y( )
BACC30BQ( )	BACC30AG( )	- - -	- - -
BACC30BS( )	BACC30AB( )CY	BACC30AB( )CY	- - -
BACC30BS( )S	BACC30AB( )SY	BACC30AB( )SY	- - -
BACC30BZ	BACC30M	BACC30R	BACC30Y( )
BACC30CP( )	BACC30CP( )	BACC30CP( )Y	- - -
BACC30CP( )S	BACC30CP( )S	BACC30CP( )SY	- - -
BACC30CQ( )CR	BACC30CQ( )CR	BACC30CQ( )CRY	- - -
BACC30CQ( )SR	BACC30CQ( )SR	BACC30CQ( )SRY	- - -
BACC30M( )	BACC30M( )	BACC30R( )	BACC30Y( )
BACC30P( )	BACC30AM( )	BACC30AM( )	BACC30AN( )
BACC30X( )	BACC30X( )	BAC30AC( )	BACC30U( )
BACC30X( )P	BACC30X( )P	BACC30AC( )P	- - -
BACC30X( )( ) FL 1	BACC30X( )( ) FL 1	- - -	- - -
BACN10MT( )	BACN10MT( )	BACN10MT( )X FL 3	BACN10MT( )X FL 3
BACN10TN( )	BACN10TN( )	- - -	- - -
BACN10XJ( )	BACN10XJ( )	BACN10XJ( )	- - -
BACN10YT( )C	BACN10YT( )C	BACN10YT( )C	- - -
BACN10YT( )CD	BACN10YT( )CD	BACN10YT( )CD	- - -
BACN10YT( )CS	BACN10YT( )CS	BACN10YT( )CS	- - -
BACN10YT( )CSA	BACN10YT( )CSA	BACN10YT( )CSA	- - -
BACN10YZ( )	BACN10XJ( )	BACN10XJ( )	- - -
BACN10WM( )	BACN10WM( )	BACN10XJ( ) FL 2	- - -
BACN10ZV( )	BACN10XJ( )	BACN10XJ( )	- - -
BACN10ZZ( )D	BACN10ZZ( )D	BACN10ZZ( )D	- - -
BACN11AJ( )	BACN10YT( )CD	BACN10YT( )CD	- - -
BACN11AJ( )CS	BACN10YT( )CS	BACN10YT( )CS	- - -
	BACN10YT( )CSA	BACN10YT( )CSA	- - -
BACN11AK( )C	BACN11AK( )C	BACN11AK( )CY	- - -
BACN11AK( )S	BACN11AK( )S	BACN11AK( )SY	- - -

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XV - ORIGINAL COLLAR OR NUT TO OVERSIZE COLLAR OR NUT CROSS REFERENCE FOR HEX DRIVE FASTENERS (Continued)

ORIGINAL COLLAR	OVERSIZE COLLARS/NUTS		
	0.0156	0.0312	0.0468
BACN11E( )	BACN11E( )	BACN11E( )	- - -
BACN11BH( )	BACN11BH( )	BACN11BH( )	- - -
BACN11BT( )	BACN10WM( )	BACN10XJ( ) FL 2	
BACC30DA( )K	BACC30DA( )K	BACC30DA( )KY	- - -
BACC30BH( )K	BACC30BH( )K	BACC30AC( )K	- - -
BACC30AB( )A	BACC30AB( )AY	BACC30AB( )AY	- - -

FL 1 Same finish code.

FL 2 Cap sealing in accordance with BAC5504 is required on the wet side of the fuel tank.

FL 3 Use next size BACW10AU washers with these oversize nuts.

TABLE XVI - HOLE SIZES FOR OVERSIZE LOCKBOLTS AND HEX-DRIVE FASTENERS (TRANSITION FIT, CLOSE REAM, CLASS I AND MODIFIED CLOSE REAM)

NOMINAL AND OVERSIZE FASTENER DIAMETER		HOLE SIZES FOR NOMINAL AND OVERSIZE REPLACEMENT FASTENERS FL 4			
		TRANSITION FIT	CLOSE REAM	CLASS 1	MODIFIED CLOSE REAM
0.1640 (-5)	MAX	0.1640	0.1645	0.1680	0.1660
	MIN	0.1610	0.1635	0.1640	0.1640
1/64 FL 1	MAX MIN	FL 3	FL 3	FL 3	N/A
1/32 FL 2	MAX	0.1900	0.1905	0.1940	0.1920
	MIN	0.1870	0.1895	0.1900	0.1900
3/64 FL 2	MAX	0.2030	0.2036	0.2070	N/A
	MIN	0.2000	0.2026	0.2030	
0.1900 (-6)	MAX	0.1900	0.1905	0.1940	0.1920
	MIN	0.1870	0.1895	0.1900	0.1900
1/64 FL 1	MAX	0.2030	0.2036	0.2070	0.2051
	MIN	0.2000	0.2026	0.2030	0.2031
1/32 FL 2	MAX	0.2190	0.2192	0.2230	0.2207
	MIN	0.2160	0.2182	0.2190	0.2187
3/64 FL 2	MAX	0.2350	0.2349	0.2390	N/A
	MIN	0.2320	0.2339	0.2350	
0.2500 (-8)	MAX	0.2500	0.2505	0.2540	0.2520
	MIN	0.2470	0.2495	0.2500	0.2500
1/64 FL 1	MAX	0.2660	0.2661	0.2700	0.2676
	MIN	0.2630	0.2651	0.2660	0.2656

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XVI - HOLE SIZES FOR OVERSIZE LOCKBOLTS AND HEX-DRIVE FASTENERS (TRANSITION FIT, CLOSE REAM, CLASS I AND MODIFIED CLOSE REAM) (Continued)

NOMINAL AND OVERSIZE FASTENER DIAMETER		HOLE SIZES FOR NOMINAL AND OVERSIZE REPLACEMENT FASTENERS FL 4			
		TRANSITION FIT	CLOSE REAM	CLASS 1	MODIFIED CLOSE REAM
1/32 FL 2	MAX	0.2810	0.2817	0.2850	0.2832
	MIN	0.2780	0.2807	0.2810	0.2812
3/64 FL 2	MAX	0.2970	0.2973	0.3010	N/A
	MIN	0.2940	0.2963	0.2970	
0.3125 (-10)	MAX	0.3130	0.3130	0.3160	0.3145
	MIN	0.3090	0.3120	0.3120	0.3125
1/64 FL 1	MAX	0.3280	0.3286	0.3320	0.3301
	MIN	0.3250	0.3276	0.3280	0.3281
1/32 FL 2	MAX	0.3440	0.3442	0.3480	0.3457
	MIN	0.3410	0.3432	0.3440	0.3437
3/64 FL 2	MAX	0.3600	0.3599	0.3640	N/A
	MIN	0.3570	0.3589	0.3600	
0.3750 (-12)	MAX	0.3750	0.3755	0.3790	0.3770
	MIN	0.3710	0.3745	0.3750	0.3750
1/64 FL 1	MAX	0.3910	0.3911	0.3950	0.3926
	MIN	0.3880	0.3901	0.3910	0.3906
1/32 FL 2	MAX	0.4060	0.4067	0.4100	0.4082
	MIN	0.4030	0.4057	0.4060	0.4062
3/64 FL 2	MAX	0.4220	0.4224	0.4260	N/A
	MIN	0.4190	0.4212	0.4220	
0.4375 (-14)	MAX	0.4380	0.4380	0.4420	0.4395
	MIN	0.4340	0.4370	0.4370	0.4375
1/64 FL 1	MAX	0.4530	0.4536	0.4580	0.4551
	MIN	0.4500	0.4526	0.4530	0.4531
1/32 FL 2	MAX	0.4690	0.4692	0.4740	0.4707
	MIN	0.4660	0.4682	0.4690	0.4687
3/64 FL 2	MAX	0.4860	0.4848	0.4900	N/A
	MIN	0.4820	0.4838	0.4860	
0.5000 (-16)	MAX	0.5000	0.5005	0.5050	0.5020
	MIN	0.4960	0.4995	0.5000	0.5000
1/64 FL 1	MAX	0.5160	0.5161	0.5210	0.5176
	MIN	0.5130	0.5151	0.5160	0.5156
1/32 FL 2	MAX	0.5310	0.5317	0.5360	0.5332
	MIN	0.5280	0.5307	0.5310	0.5312
3/64 FL 2	MAX	0.5470	0.5474	0.5510	N/A
	MIN	0.5440	0.5464	0.5470	

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XVI - HOLE SIZES FOR OVERSIZE LOCKBOLTS AND HEX-DRIVE FASTENERS (TRANSITION FIT, CLOSE REAM, CLASS I AND MODIFIED CLOSE REAM) (Continued)

NOMINAL AND OVERSIZE FASTENER DIAMETER		HOLE SIZES FOR NOMINAL AND OVERSIZE REPLACEMENT FASTENERS FL 4			
		TRANSITION FIT	CLOSE REAM	CLASS 1	MODIFIED CLOSE REAM
0.5625 (-18)	MAX	0.5630	0.5630	0.5670	0.5640
	MIN	0.5590	0.5620	0.5620	0.5620
1/64 FL 1	MAX	0.5790	0.5786	0.5830	0.5796
	MIN	0.5750	0.5776	0.5780	0.5776
1/32 FL 2	MAX	0.5950	0.5942	0.5980	0.5952
	MIN	0.5910	0.5932	0.5930	0.5932
3/64 FL 2	MAX	0.6090	0.6099	0.6130	N/A
	MIN	0.6060	0.6089	0.6090	
0.6250 (-20)	MAX	0.6250	0.6255	0.6300	0.6270
	MIN	0.6210	0.6245	0.6250	0.6250
1/64 FL 1	MAX	0.6410	0.6411	0.6460	0.6426
	MIN	0.6370	0.6401	0.6410	0.6406
1/32 FL 2	MAX	0.6570	0.6567	0.6610	0.6582
	MIN	0.6530	0.6557	0.6560	0.6562
3/64 FL 2	MAX	0.6720	0.6724	0.6760	N/A
	MIN	0.6690	0.6714	0.6720	

FL 1 Quality Assurance verification required

FL 2 Engineering approval and Quality Assurance verification are required if edge margin is less than two diameters, whereas only Quality Assurance verification is required if the edge margin is twice the diameter of the original fastener, or greater.

FL 3 Use (-6) nominal size as 0.0312 inch oversize for (-5). FL 2 applies. There is no 0.0156 inch oversize for (-5).

FL 4 Replacement fastener hole sizes shall be selected in accordance with original hole callout on drawing.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

TABLE XVII - HOLE SIZES FOR OVERSIZE LOCKBOLTS AND HEX DRIVE FASTENERS (INTERFERENCE FIT)

ORIGINAL NOMINAL BOLT AND OVERSIZE FL 2	HOLE SIZE-ORIGINAL AND OVERSIZE INTERFERENCE FIT FL 1				ORIGINAL NOMINAL BOLT AND OVERSIZE FL 2	HOLE SIZE-ORIGINAL AND OVERSIZE INTERFERENCE FIT FL 1			
	0.0000 - 0.0030	**0.0005 - 0.0035**	**0.0005 - 0.0045**	**0.0010 - 0.0040**		**0.0000 - 0.0030**	**0.0005 - 0.0035**	**0.0005 - 0.0045**	**0.0010 - 0.0040**
-5	0.1605 0.1625	0.1600 0.1620	0.1590 0.1620	0.1595 0.1615	-14	0.4340 0.4360	0.4335 0.4355	0.4325 0.4355	0.4330 0.4350
1/64	FL 3	FL 3	FL 3	FL 3	1/64	0.4496 0.4516	0.4491 0.4511	0.4481 0.4511	0.4486 0.4506
1/32	FL 3	FL 3	FL 3	FL 3	1/32	0.4652 0.4672	0.4647 0.4667	0.4637 0.4667	0.4642 0.4662
-6	0.1865 0.1885	0.1860 0.1880	0.1850 0.1880	0.1855 0.1875	-16	0.4965 0.4985	0.4960 0.4980	0.4950 0.4980	0.4955 0.4975
1/64	0.1996 0.2016	0.1991 0.2011	0.1981 0.2011	0.1986 0.2006	1/64	0.5121 0.5141	0.5116 0.5136	0.5106 0.5136	0.5111 0.5131
1/32	0.2152 0.2172	0.2147 0.2167	0.2137 0.2167	0.2142 0.2162	1/32	0.5277 0.5297	0.5272 0.5292	0.5262 0.5292	0.5267 0.5287
-8	0.2465 0.2485	0.2460 0.2480	0.2450 0.2480	0.2455 0.2475	-18	0.5585 0.5605	0.5580 0.5600	0.5570 0.5600	0.5575 0.5595
1/64	0.2621 0.2641	0.2616 0.2636	0.2606 0.2636	0.2611 0.2631	1/64	0.5741 0.5761	0.5736 0.5766	0.5726 0.5756	0.5731 0.5751
1/32	0.2777 0.2797	0.2772 0.2792	0.2762 0.2792	0.2767 0.2787	1/32	0.5897 0.5917	0.5892 0.5912	0.5882 0.5912	0.5887 0.5907
-10	0.3090 0.3110	0.3085 0.3105	0.3075 0.3105	0.3080 0.3100	-20	0.6210 0.6230	0.6205 0.6225	0.6195 0.6225	0.6200 0.6220
1/64	0.3246 0.3266	0.3241 0.3261	0.3231 0.3261	0.3236 0.3256	1/64	0.6366 0.6386	0.6361 0.6381	0.6351 0.6381	0.6356 0.6376
1/32	0.3402 0.3422	0.3397 0.3417	0.3387 0.3417	0.3392 0.3412	1/32	0.6522 0.6542	0.6517 0.6537	0.6507 0.6537	0.6512 0.6532
-12	0.3715 0.3735	0.3710 0.3730	0.3700 0.3730	0.3705 0.3725	-24	0.7460 0.7480	0.7455 0.7475	0.7445 0.7475	0.7450 0.7470
1/64	0.3871 0.3891	0.3866 0.3886	0.3856 0.3886	0.3861 0.3881	1/64	0.7616 0.7636	0.7611 0.7631	0.7601 0.7631	0.7606 0.7626
1/32	0.4027 0.4047	0.4022 0.4042	0.4012 0.4042	0.4017 0.4037	1/32	0.7772 0.7792	0.7767 0.7787	0.7757 0.7787	0.7762 0.7782

FL 1 For proper installation hole diameter refer to Engineering drawing for the as called out initial hole size.

FL 2 First oversize with edge margin 2.5 times the diameter of the original hole size or equivalent to Engineering drawing spacing/edge distance, Quality Assurance verification is required.

Copyright © Boeing. All rights reserved. See Cover Page.

8.2.11 FASTENER REPLACEMENT-OVERSIZE (Continued)

First oversize with edge margin less than 2.5 times the diameter of the original hole size or less than the Engineering drawing spacing/edge distance, Engineering approval and Quality Assurance verification are required.

Second oversize and larger, Engineering approval and Quality Assurance verification are required.

FL 3 Use -6 nominal size as 1/32 inch oversize for -5. FL 2 applies. There is no 1/64 inch oversize for -5.

8.3 REWORK

a. Document rework as required by the applicable quality assurance provisions.

b. For hex-drive bolts, re-seating of the pin is permitted providing that the collar is replaced or the nut is re-tightened in accordance with Section 8.2.9.f. or Section 8.2.9.o. as applicable, and providing that replacement of the bolt is not required by Section 8.2.9.m. and Section 8.2.9.n.

c. For lockbolts and hex-drive fasteners with machine swaged lockbolt collars, additional driving is not permitted. Lockbolts shall be replaced.

d. For hex-drive bolts with machine swaged lockbolt collars, if the machine drills and installs the bolt but the cycle is interrupted prior to collar swage, contact Engineering liason.

e. For hex-drive bolts with machine swaged lockbolt collars, oversize rework shall be in accordance with Section 8.2.11.

f. When replacing fasteners, any damage to the holes shall be reported to Quality Assurance prior to rework to the next oversize.

g. Hole sizes for oversize hex-drive fasteners and lockbolts shall be as listed in Table XVI and Table XVII unless otherwise called out in the Engineering drawing.

9 MAINTENANCE CONTROL

9.1 TOOLS

9.1.1 HI-SHEAR RIVET AND LOCKBOLT TOOLS

Collar swage dies shall be inspected at 6-month maximum intervals. Dies with swage cavities that are out of tolerance shall not be used for production.

9.1.2 HI-SHEAR RIVET TOOLS

Collar swage dies shall conform to the applicable ST1109 tool, or equivalent.

9.1.3 LOCKBOLT COLLAR SWAGE DIE

a. Lockbolt collar swage die cavity dimensions shall conform to configurations called out in Section 8.2.7, Table V, Table VI, and Figure 11.

Copyright © Boeing. All rights reserved. See Cover Page.

9.1.3 LOCKBOLT COLLAR SWAGE DIE (Continued)

b. Tooling Services or Quality Assurance must verify the collar swage dies at a maximum interval of 6 months. Use the appropriate Go-No-Go gage to verify the 'A' dimensions.

c. Each shop/site shall have a documented process for monitoring the condition of the dies to prevent use of out-of-tolerance or damaged dies.

CAUTION It is recommended that swage dies be cleaned daily. Cured sealant is difficult to remove. The buildup of cured sealant may damage the swage die and result in unreliable fastener installations.

9.1.4 HEX DRIVE BOLT TOOLS

a. Internal hexagonal tools shall be inspected if fastener installation difficulties occur. Tools with hexagonal recesses which are out of tolerance shall not be used for production. Use gages in accordance with ANSI/ASME B107.17M

b. External hexagonal tools (hexagon keys) shall be inspected if fastener installation difficulties occur. Tools out of tolerance shall not be used for production.

c. Internal hexagonal tool recess configuration shall conform to FED-STD-H28, Appendix A10.

d. External hexagonal tool (hexagon keys) configuration shall conform to ANSI/ASME B18.3.

e. Use of impact wrench is prohibited.

f. Tools with ratcheting clutches or power tools designated for torquing off collars shall not be used to apply or control installation torques of nuts.

9.1.5 HEX-DRIVE BOLT - LOCKBOLT COLLAR TOOLS

Swage dies used for this process shall comply with Figure 10.

9.2 GAGES

a. Tools and gages that are used to verify the requirements of Section 11 must comply with the requirements specified in AS9100.

b. Protrusion gages, feeler stock, or grip gauges which are out of tolerance shall not be used.

9.3 AUTOMATIC FASTENING MACHINES

a. Automatic fastening machines shall be subject to an active and documented preventative maintenance plan. This plan shall, as a minimum, provide for periodic inspection and maintenance of the following machine functions:

(1) Hydraulic system performance

(2) Drill spindle speeds and feeds

(3) Upper ram to lower ram alignment

Copyright © Boeing. All rights reserved. See Cover Page.

9.3 AUTOMATIC FASTENING MACHINES (Continued)

(4) Drill spindle to upper ram alignment

(5) Drill spindle to lower ram alignment

(6) Clamping and upset or installation pressures

(7) Drill spindle run out

b. The preventative maintenance plan shall be related to machine operating cycle. The contractor shall make his preventative maintenance plan available to Boeing for review and approval.

c. The following automatic fastening machines which install fasteners through fuselage skin shall meet the requirements of D6-56617

(1) Automatic fastening machines which install protruding and flush head hex-drive bolts and lockbolts in fuselage skins, or install lockbolt collars on hex drive fasteners.

(2) Automatic fastening machines which install protruding and flush head stump type lockbolts in all fuselage structure.

9.4 TORQUE TOOL REQUIREMENTS

Refer to BSS7083 for the calibration and certification requirements for torque tools.

10 QUALITY CONTROL

a. Assure that the requirements of this specification are met by monitoring the process and examining the end-items in accordance with established quality assurance provisions.

b. The preferred method is to employ in-process monitoring using SPC analysis of variable measurement data, as described in Section 10.1 below. The other methods described in Section 10.2 through Section 10.4 are acceptable alternatives. A combination of two or more methods may also be used. The sampling requirements contained in this document should not be construed as permitting acceptance of defective parts. It is expected that all parts be 100 percent conforming to all engineering requirements.

c. Verify that only certified and appropriately maintained tools and gauges are used for Quality Assurance inspection of the process.

d. Quality Assurance shall verify that automatic hydraulic squeeze fastening equipment used to install fasteners to the requirement of this specification is certified in accordance with Section 9.3.c.

e. Verify that torque tools used to perform this process are maintained and certified, as required. Refer to BSS7083 for requirements.

f. The reliability requirements tabulated in Table XVIII define the level of verification required to demonstrate this conformance.

Copyright © Boeing. All rights reserved. See Cover Page.

10 QUALITY CONTROL (Continued)

TABLE XVIII - INSPECTION RELIABILITY REQUIREMENT

CHARACTERISTIC	PERCENT
1. Holes a. Diameter b. Radius or chamfer c. Surface Characteristics d. Angularity e. Countersinking and dimpling	98
2. Use of proper washer 3. Flushness and head dishing 4. Bolt Protrusion and Collar (Swage or Torque Off) 5. Gaps and part separation	97
6. Nut torque	97
7. All other Engineering requirements	90

g. Unless specifically stated by the sampling plan, all acceptance sampling rules should be interpreted as being based on a single inspection of the feature or characteristic of interest. If a measurement or test fails the requirement, which leads to product or lot rejection, then any engineering authorized re-work should be viewed as corrective action. Such re-work does not change any statistics associated with sampling rules. Re-work or part replacement is tied to corrective action to restore the lot or product set.

10.1 IN-PROCESS VERIFICATION - VARIABLE MEASUREMENTS

a. Each operator should implement statistical process controls (SPC) in accordance with BSS7286 for key characteristics and key process parameters. If the operator uses SPC for any characteristic, and meets the requirements of BSS7286, then end-item sampling in accordance with Table XIX is not required for those characteristics. The appropriate sampling frequency for the SPC will cause a minimum of one applicable fastener installation to be inspected for every eight made, as an average, and at least one per month regardless of rate.

TABLE XIX - SAMPLE SIZE REQUIREMENTS, FL 1

LOT SIZE	INSPECTION RELIABILITY REQUIREMENT			
	90	97	98	99
	SAMPLE SIZE			
Up to 17	6	All	All	All
18 to 25	6	17	All	All
26 to 50	6	19	25	All
51 to 102	6	20	29	50
103 to 106	6	21	30	51
107 to 111	6	21	30	52
112 to 116	6	21	30	53
117 to 123	6	21	30	54

Copyright © Boeing. All rights reserved. See Cover Page.

10.1 IN-PROCESS VERIFICATION - VARIABLE MEASUREMENTS (Continued)

TABLE XIX - SAMPLE SIZE REQUIREMENTS, FL 1
(Continued)

LOT SIZE	INSPECTION RELIABILITY REQUIREMENT			
	90	**97**	**98**	**99**
	SAMPLE SIZE			
124 to 130	6	21	31	55
131 to 140	6	21	31	56
141 to 155	6	21	31	57
156 to 200	6	22	32	58
201 to 214	7	22	32	59
215 to 233	7	22	32	60
234 to 270	7	22	32	61
271 to 323	7	22	33	62
324 to 403	7	22	33	63
404 to 502	7	22	33	64
503 to 633	7	22	33	65
634 to 917	7	22	34	66
918 to 1555	7	23	34	67
1556 to 5000	7	23	34	68
5001 and up	7	23	34	69

FL 1 This table applies only to the sampling of attribute measurements.

b. The sample statistic Cpk shall be at least 1.33 or yield at least 90 percent confidence that the production process is operating within the reliability requirements of Table XVIII.

10.2 IN-PROCESS VERIFICATION - ATTRIBUTE MEASUREMENTS

a. Quality Assurance shall inspect end-items to ensure the process is controlled in accordance with Section 10.2.d. below.

b. Lot definition and inspection procedures shall be in accordance with the operating division's Quality Assurance provisions.

c. An equivalent Quality Assurance approved plan which provides assurance that the requirements of this process specification are being met may be used in place of sampling in accordance with Table XIX.

d. Samples sizes are found in Table XIX. If any units are found defective, then the entire lot is rejected.

10.3 END-ITEM VERIFICATION - VARIABLE MEASUREMENTS

End-item verification using variables measurements is acceptable for all characteristics which are end-item inspectable.

Copyright © Boeing. All rights reserved. See Cover Page.

10.4 END-ITEM VERIFICATION - ATTRIBUTE MEASUREMENTS

End-item verification using inspection by attributes is acceptable for all characteristics that are end-item inspectable.

11 REQUIREMENTS

11.1 HOLES/INSTALLATION

a. Hole preparation and installation of permanent fasteners shall be carried out in accordance with Section 8.1.

b. All fastener hole characteristics are Engineering requirements.

c. Hole sizes shall be in accordance with Table I, unless otherwise specified on the Engineering drawing. Oversizing the non-conforming holes is considered a rework and shall be documented in accordance with Section 8.3.

d. Hole sizes are key characteristics, unless otherwise specified, if statistical process controls (SPC) method is used.

NOTE: Fastener holes have the following characteristics: (1) size, for example, diameter, (2) fillet relief (to prevent interference with bolt head-to-shank fillet radius), (3) surface texture, for example, surface roughness, (4) angularity, for example, perpendicularity, (5) countersink, dimple, and (6) geometric properties, for example, orientation and alignment of the countersink to the hole. Tests have shown that hole characteristics have a major impact on the dynamic durability of the structure, on the ability of the fastener to transfer design loads across the joint faying surface, and on the service life of the fastener.

11.2 GENERAL REQUIREMENTS FOR INSTALLED FASTENERS

a. Collars that are cracked after installation shall be replaced. Visible circumferential laps on the top of any installed aluminum lockbolt collar are acceptable provided material has not been removed or separated from the collar during installation (see Figure 14). Removal of material from collar after installation is complete is not allowed and shall be cause for rejection.

Copyright © Boeing. All rights reserved. See Cover Page.

11.2 GENERAL REQUIREMENTS FOR INSTALLED FASTENERS (Continued)

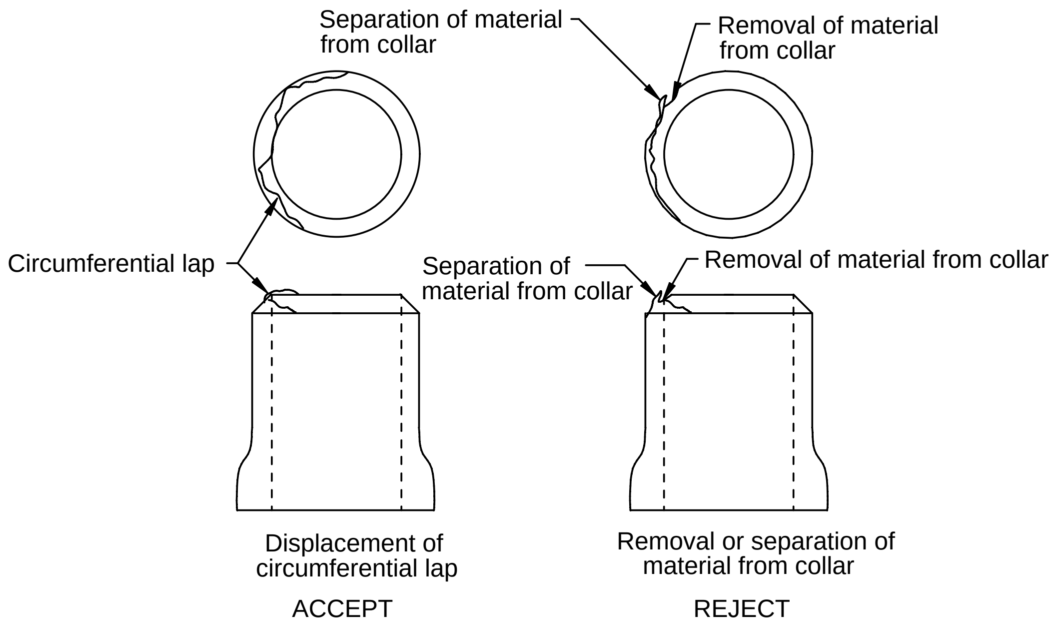


FIGURE 14 - CIRCUMFERENTIAL LAPS

b. When a washer is used under the collar, pin protrusion shall be checked from the top of the washer with the specified gage, except when washers are used for counterbore substitution.

c. Unless otherwise stated on the applicable aero-flushness drawing or installation drawing, or elsewhere in this specification, flushness limits for flat and domed flush head fasteners shall be +0.010/-0.005.

(1) Those fasteners that are overcoated with BMS5-62, BMS8-70, BMS8-78, or BMS8-103 shall be flush within +0.015/-0.005 inch.

(2) Flushness limits for domed head flush fasteners (BACB30XT, BACB30XU and BACB30YP), installed on the exterior surface, shall be +0.0085/+0.0025 inch unless otherwise stated on the applicable aero-flushness drawing or installation drawing.

(3) All flat and domed head flush hex drive bolt and lockbolt installations where the head of the fastener is covered by another piece of structure placed in direct contact upon assembly shall be flush within +0.000/–0.005 inch unless otherwise noted on the applicable installation drawing.

(4) All flat and domed flush head hex drive fasteners and lockbolts which are installed below flush shall use an installation process or tool designed to ensure that the fastener is fully seated prior to the installation of the collar or nut in accordance with Section 8.2.1.a.

(5) See Figure 16 for explanation of flushness.

(6) If flushness limits are not met:

Copyright © Boeing. All rights reserved. See Cover Page.

11.2 GENERAL REQUIREMENTS FOR INSTALLED FASTENERS (Continued)

(a) For hex-drive bolts with threaded collars or nuts, additional driving of pin is permitted provided the collar is replaced or the nut is re-tightened in accordance with Section 8.2.9.d. or Section 8.2.9.f. as applicable and replacement of bolt is not required by Section 8.2.9.l. and Section 8.2.9.m.

(b) For lockbolts additional driving is not permitted. The fastener shall be replaced. For hex-drive fasteners with machine swaged collars, additional driving is not permitted. The installation may be reworked in accordance with Section 8.3.

d. Head dishing for flush head fasteners shall not exceed the limits of Figure 15 and Table XX. Head dishing shall be smooth and uniform. Head dishing that shows a distinct ring is not acceptable.

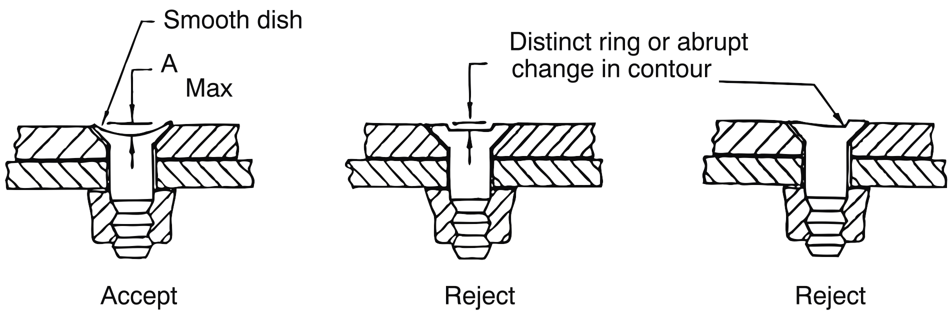


FIGURE 15 - FASTENER HEAD DISHING

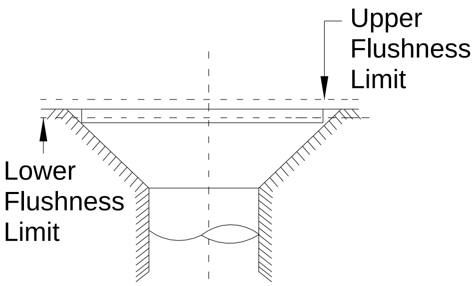
TABLE XX - FASTENER HEAD DISH LIMITS

APPLICABLE FASTENER TYPE	FASTENER MATERIAL	NOMINAL DIAMETER	A MAX
Pull-Type Shear Head Lockbolts	Aluminum and Alloy Steel	0.190	0.012
		0.250	0.012
		0.312	0.010
		0.375 and larger	0.007
	A-286 CRES	ALL	0.008
	Titanium	ALL	0.004
Pull-Type Tension Head Lockbolts	Titanium	ALL	0.004
All Others	ALL	ALL	0.004

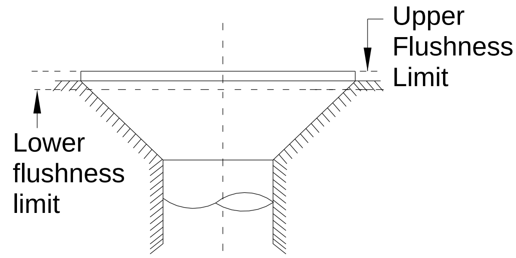
e. Discoloration of the corners of the wrenching features of aluminum nuts is acceptable. There shall be no permanent deformation of the wrenching feature that interferes with the use of box, socket, or open end wrenches, or which prevents further tightening or the removal of the nut.

Copyright © Boeing. All rights reserved. See Cover Page.

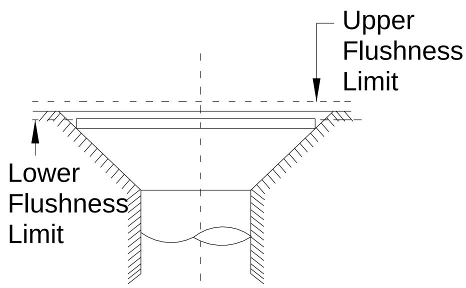
11.2 GENERAL REQUIREMENTS FOR INSTALLED FASTENERS (Continued)



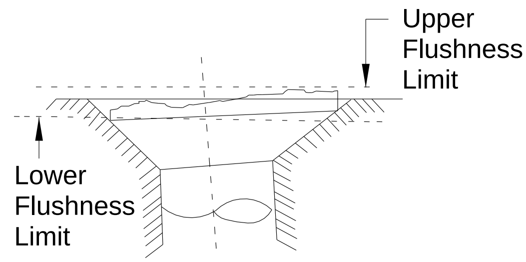
Fastener exactly flush



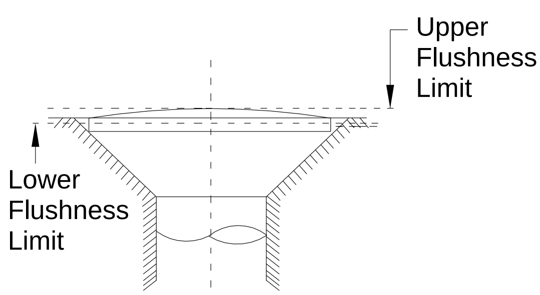
Fastener exactly at upper flushness limit
fastener just meets flushness limit



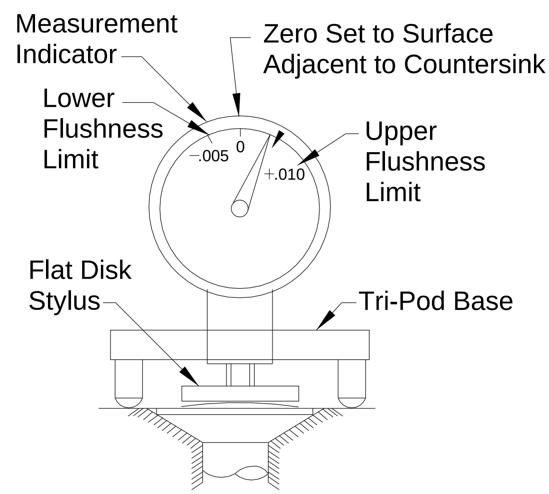
Fastener exactly at lower flushness limit.
Fastener just meets flushness limit.



Fastener hole not perpendicular to surface. Top of fastener head not perpendicular to shank. Top of fastener head not flat. Fastener flushness just meets flushness limits.



Fastener with crown on top of head.
Fastener flushness just meets flushness limit.



Measurement indicator and flat disk stylus for determining flushness values
(This method also applies to flush head fasteners without the dome)

FIGURE 16 - EXPLANATION OF FLUSHNESS

Copyright © Boeing. All rights reserved. See Cover Page.

11.2.1 GAP EVALUATION

a. There shall be no measurable gap under the collar or nut of an installed lockbolt or Hi-Shear rivet or under a self-aligning collar or self-aligning nut of an installed hex drive bolt.

b. Collar or nuts on hex drive bolts shall seat in at least one location on the periphery of the collar or nut, and, feeler stock in accordance with Table XXI, cannot enter between the structure and collar or nut, as shown in Figure 18(a).

Exception: When collars or nuts on hex–drive bolts are installed in tube assemblies, the values of Table XXI do not apply. See Figure 18(e) for requirements.

c. The heads of protruding head fasteners shall seat in at least one location on the periphery of the head and feeler stock in accordance with Table XXI, cannot enter between head and structure, as shown in Figure 18(b).

Exception: When collars or nuts on hex–drive bolts are installed in tube assemblies, the values of Table XXI do not apply. See Figure 18(e) for requirements.

d. The heads of all flush head fasteners shall seat in at least one location on the periphery of the head and the remaining gap shall be such that a 0.0030 ± 0.0005 inch feeler stock cannot enter between fastener head and countersink, as shown in Figure 18(d).

e. When fasteners are installed with wet sealant in accordance with the drawing and or BAC specifications, care should be used when checking for gaps. If sealant is damaged, fastener replacement is required. If the gap inspection damages wet sealant touch up the sealant as required. If the gap inspection damages cured sealant the fastener must be removed and replaced.

f. Sum of gaps between, structure and washer(s) and collar or nut shall be less than the values shown in Table XXI.

g. Sum of gaps between structure, washer(s) and the heads of protruding head fasteners shall be less than the values shown in Table XXI.

h. Feeler stock to be used for gap inspection shall conform to the dimensions as shown in Figure 17.

i. For lockbolts installed by pull-in process in accordance with Section 8.2.8, the head gap requirements apply before collar installation.

j. For flush head lockbolts installed by pull-in process Section 8.2.8, the gap requirements in accordance with Section 8.2.8.f.

k. There shall be no measurable gap under a swaged collar installed on a hex-drive fastener.

Copyright © Boeing. All rights reserved. See Cover Page.

11.2.1 GAP EVALUATION (Continued)

TABLE XXI - FEELER STOCK SIZE FOR EVALUATING GAPS UNDER COLLARS, NUTS AND PROTRUDING HEADS OF LOCKBOLTS AND HEX-DRIVE FASTENERS

NOMINAL FASTENER DIAMETER (INCH)	FEELER STOCK THICKNESS ± 0.0005
0.164	0.0040
0.190	0.0050
0.250	0.0060
0.313	0.0070
0.375	0.0080
0.438	0.0090
0.500	0.0100
0.563	0.0110
0.625	0.0130
0.750	0.0150
0.875	0.0170
1.000	0.0190

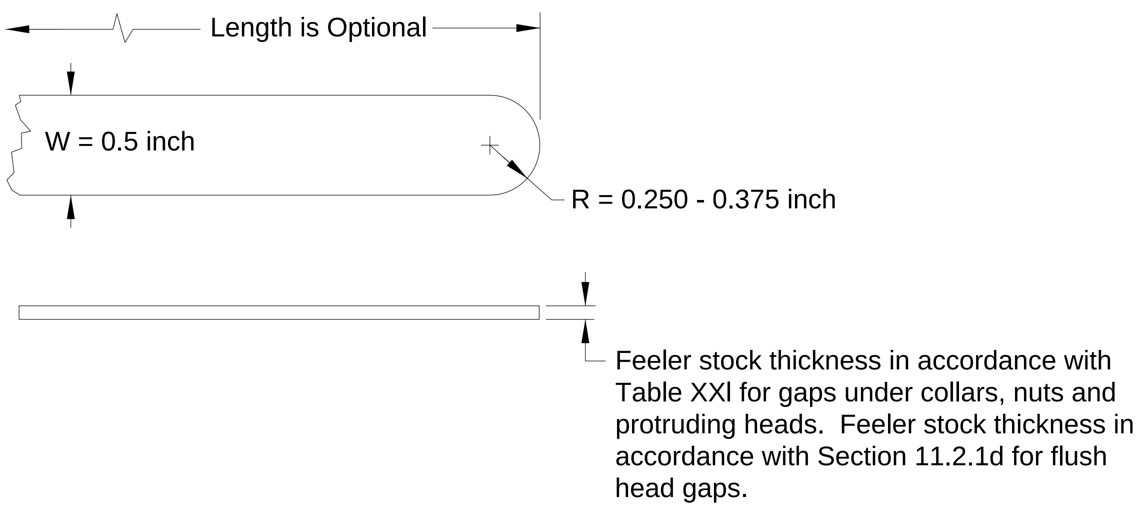
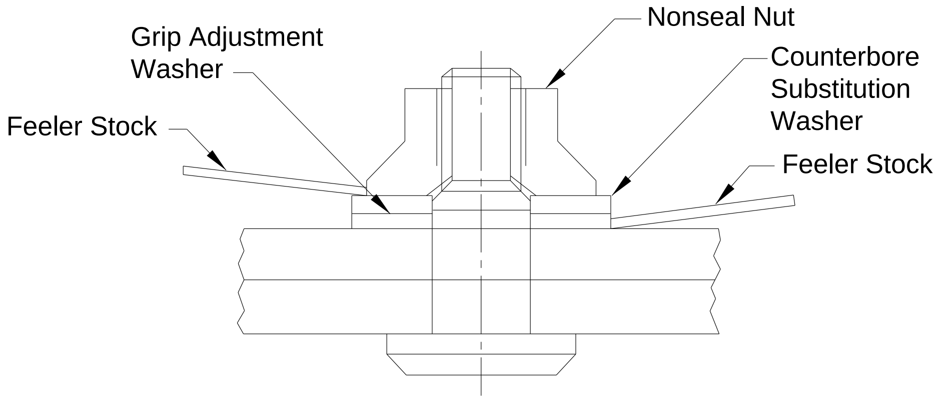


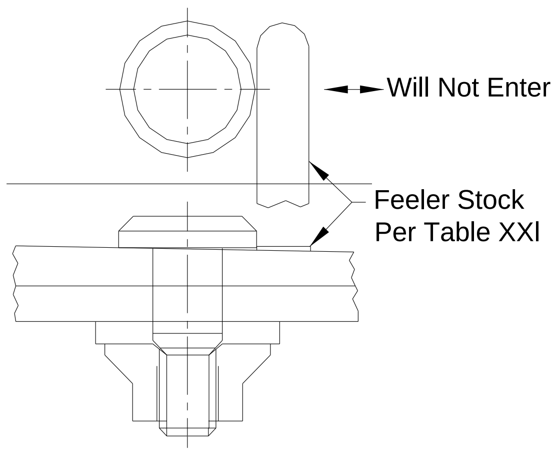
FIGURE 17 - GAP INSPECTION FEELER STOCK

Copyright © Boeing. All rights reserved. See Cover Page.

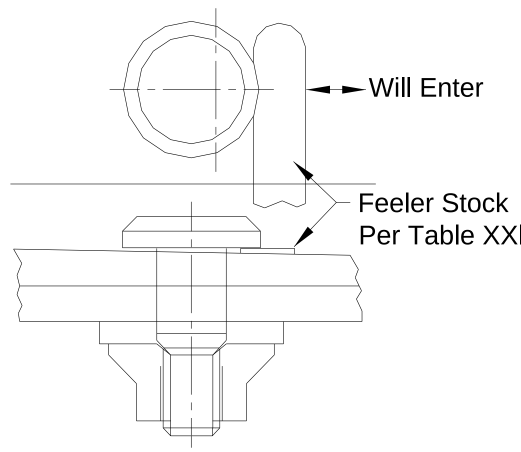
11.2.1 GAP EVALUATION (Continued)



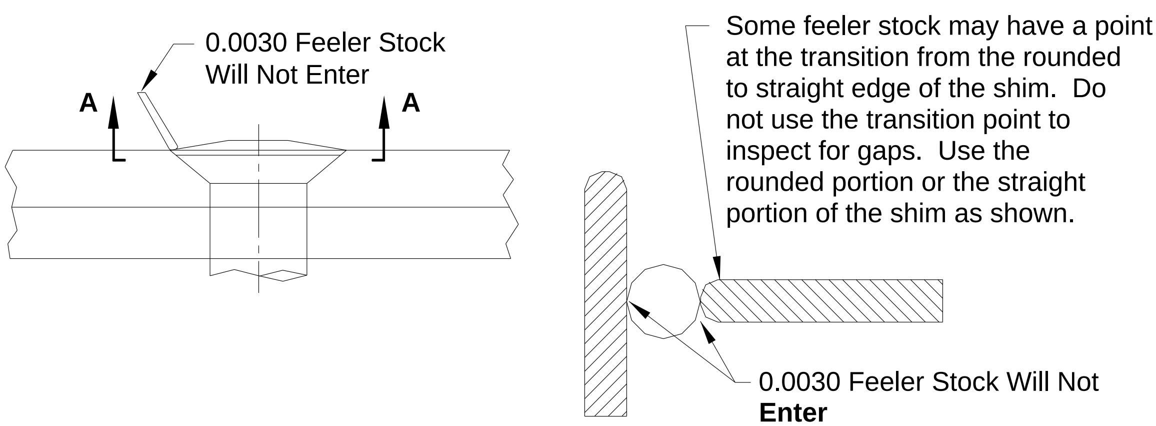
a) CHECKING FOR GAPS UNDER NUTS/COLLARS (HEX-DRIVE FASTENERS ONLY)



b) ACCEPTABLE HEAD GAP



c) REJECTABLE HEAD GAP



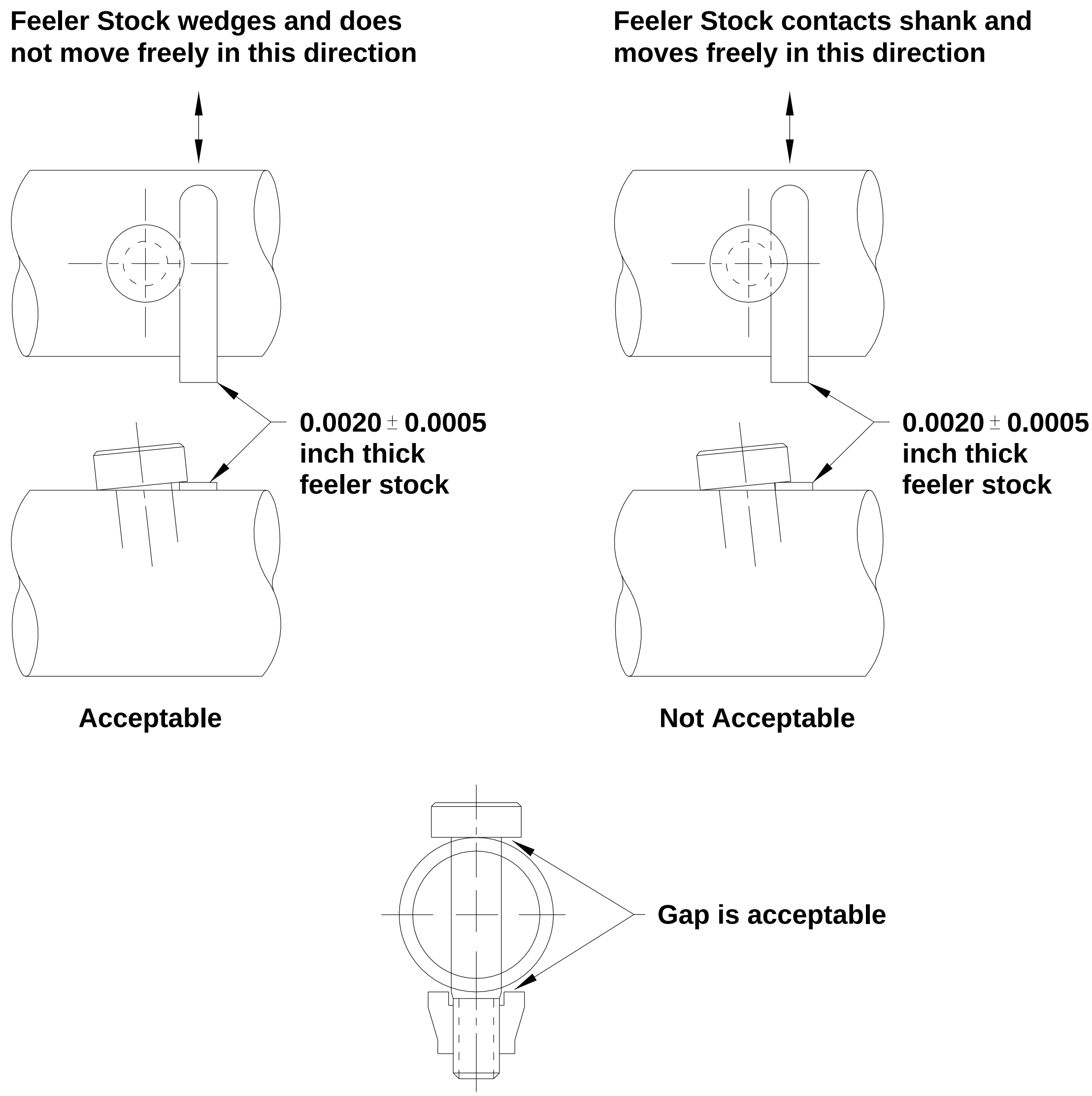
VIEW A-A

d) FLUSH HEAD GAP CHECKING METHOD

FIGURE 18 - GAP CHECKING METHODS

Copyright © Boeing. All rights reserved. See Cover Page.

11.2.1 GAP EVALUATION (Continued)



e) PROTRUDING HEAD GAP CHECKING METHOD, TUBE ASSEMBLIES

FIGURE 19 - GAP CHECKING METHODS (continued)

11.2.2 PART SEPARATION

After fastener installation, there shall be no measurable gap at the fastener shank.

Copyright © Boeing. All rights reserved. See Cover Page.

11.3 INSTALLED REQUIREMENTS

11.3.1 HI-SHEAR RIVETS

a. The pin protrusion H shall be within the limits specified in Table XXII. When the collar is driven against a sloping surface, the pin protrusion is to be measured at the maximum and minimum points. Use of an aluminum washer to facilitate swaging (as illustrated in Figure 19 and with pin protrusion limits as shown in Table XXII) is acceptable.

TABLE XXII - PIN PROTRUSION LIMITS

NOMINAL HI-SHEAR RIVET DIAMETER (INCH)	PIN PROTRUSION	
	H MINIMUM	H MAXIMUM
3/16	0.115	0.199
1/4	0.155	0.239
5/16	0.197	0.281
3/8	0.238	0.322

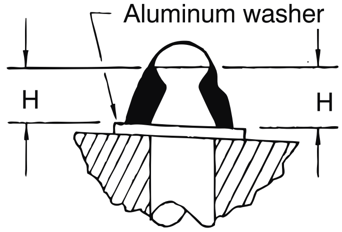


FIGURE 19 - PIN PROTRUSION

b. NAS179 (2117-T4) aluminum alloy collars shall be swaged down all around so that they trim off flush with the edge of the pin. A slight burr or ragged broken edge at the trim off point is acceptable as illustrated in Figure 20 (a).

c. A shoulder at the base of the collar as shown in Figure 20(a) is acceptable.

d. A gap between the cutting edge of the pin and the collar (such as the one shown in Figure 20(b) is usually caused by use of too long a pin and is not acceptable.

e. NAS528 (2024-T4) aluminum alloy collars shall be swaged down all around so that they trim off within the limits shown in Figure 21 and Table XXIII.

f. Unless otherwise specified on the drawing, flushness limits of flush head Hi-Shear rivets shall be +0.010/-0.000 inch.

TABLE XXIII - ECCENTRIC TRIMOFF

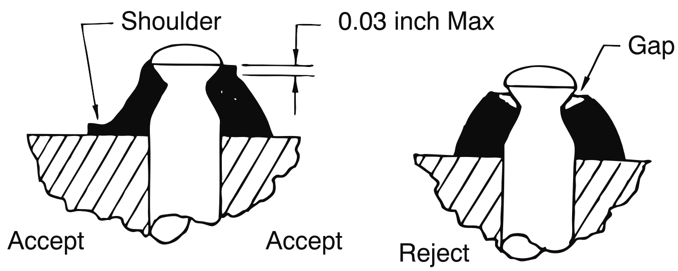
NOMINAL HI-SHEAR RIVET DIAMETER (INCH)	PIN PROTRUSION
	G MINIMUM
0.190	0.015
0.250	0.020
0.313	0.025

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.1 HI-SHEAR RIVETS (Continued)

TABLE XXIII - ECCENTRIC TRIMOFF
(Continued)

NOMINAL HI-SHEAR RIVET DIAMETER (INCH)	PIN PROTRUSION G MINIMUM
0.375	0.030



(a) Collar driving limit (b) Pin too long

FIGURE 20 - HI-SHEAR PIN COLLARS

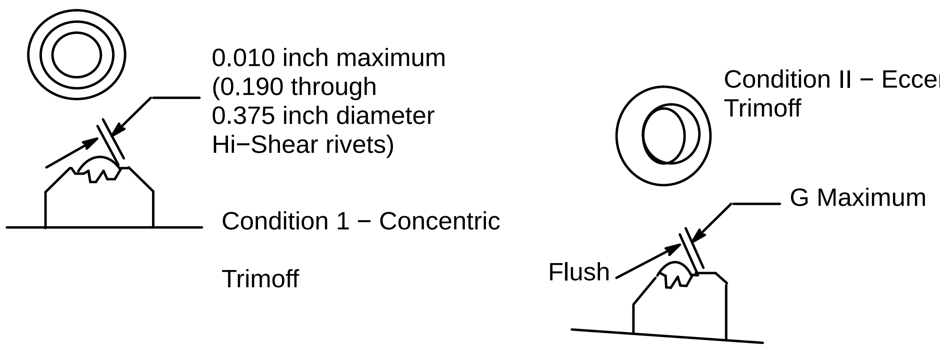


FIGURE 21 - INSTALLED HI-SHEAR RIVET TRIM-OFF

11.3.2 LOCKBOLTS

a. When collars are installed against a sloping surface, the pin protrusion shall be checked with the plane of the protrusion gage perpendicular to the slope of the surface.

b. Lockbolt protrusion and collar swage shall meet the requirements of Table XXIV and Table XXVI and the figure noted therein.

c. When washers are used under collars, pin protrusion shall be checked from the top of the washer(s).

d. Pin protrusion for light weight pull-type lockbolts with sealant escape grooves (BACB30VN( )HK and BACB30XT( )HK) shall meet the requirements of Table XXV and Figure 22 noted therein.

NOTE: Installations with lockbolts with the sealant escape grooves are not considered fluid tight.

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)

TABLE XXIV - LIGHTWEIGHT LOCKBOLT INSPECTION DIMENSION (FOR LOCKBOLTS WITHOUT SEALANT ESCAPE GROOVES) FL 1

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y		Z	R	T MAX		FIG. NO.	GAGE NO.
			TOUCH GO ± 0.001	TOUCH NO GO	(REF)	(REF)	TOUCH GO	TOUCH NO GO		
0.1640	BACB30VM5 BACB30VN5	BACC30BK5 (2024-T4)	0.131	0.237	0.124	0.234	0.039	0.112	Figure 22	HG 110-05
	BACB30VP5 BACB30VR5	BACC30BK5 (2024-T4)	0.141	0.248	N/A	0.234	0.039	0.113	Figure 23	HG 113-05
0.1900	BACB30VM6 BACB30VN6 BACB30XT6	BACC30BK6 (2024-T4)	0.134	0.250	0.152	0.262	0.031	0.113	Figure 22	HG 110-06
	BACB30VP6 BACB30VR6 BACB30XU6	BACC30BK6 (2024-T4)	0.152	0.258	N/A	0.262	0.043	0.123	Figure 23	HG 113-06
0.2500	BACB30VM8 BACB30VN8 BACB30XT8	BACC30BK8 (2024-T4)	0.193	0.301	0.209	0.344	0.058	0.139	Figure 22	HG 110-08
	BACB30VP8 BACB30VR8 BACB30XU8	BACC30BK8 (2024-T4)	0.213	0.319	N/A	0.344	0.070	0.143	Figure 23	HG 113-08
0.3125	BACB30VM10 BACB30VN10 BACB30XT10	BACC30BK10 (2024-T4)	0.252	0.367	0.259	0.436	0.072	0.145	Figure 22	HG-110 -10
	BACB30VP10 BACB30VR10 BACB30XU10	BACC30BK10 (2024-T4)	0.279	0.385	N/A	0.436	0.077	0.146	Figure 23	HG 113-10
0.3750	BACB30VM12 BACB30VN12 BACB30XT12	BACC30BK12 (2024-T4)	0.304	0.419	0.322	0.512	0.103	0.172	Figure 22	HG 110-12
	BACB30VP12 BACB30VR12 BACB30XU12	BACC30BK12 (2024-T4)	0.329	0.435	N/A	0.512	0.109	0.177	Figure 23	HG 113-12

FL 1 HG110 and HG113 gages have yellow bands painted in the middle. These are different from the ones with green bands used for lockbolts with sealant escape grooves.

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)

TABLE XXV - LIGHT WEIGHT LOCKBOLT INSPECTION DIMENSION (FOR LOCKBOLTS WITH SEALANT ESCAPE GROOVES) FL 1

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y		Z (REF)	R (REF)	T MAX		FIG. NO.	GAGE NO. FL 1
			TOUCH GO	TOUCH NO GO			TOUCH GO	TOUCH NO GO		
0.1900	BACB30VN6HK BACB30XT6HK	BACC30BK6 (2024-T42)	0.136	0.230	0.152	0.262	0.031	0.113	Figure 22	HG 160-06
0.2500	BACB30VN8HK BACB30XT8HK	BACC30BK8 (2024-T42)	0.194	0.288	0.209	0.344	0.058	0.139	Figure 22	HG 160-08
0.3125	BACB30VN10HK BACB30XT10HK	BACC30BK10 (2024-T42)	0.253	0.347	0.259	0.436	0.072	0.145	Figure 22	HG 160-10
0.3750	BACB30VN12HK BACB30XT12HK	BACC30BK12 (2024-T42)	0.305	0.399	0.322	0.512	0.103	0.172	Figure 22	HG 160-12

FL 1 HG160 gages have a green band painted in the middle.

TABLE XXVI - STANDARD LOCKBOLT INSPECTION DIMENSION

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y	Z	R	T	FIG. NO.	GAGE NO.
0.1640	BACB30GP5 BACB30GQ5 NAS1525 NAS1535 (7075-T6)	NAS1080D05 (6061-T7)	0.230 0.167	0.136	0.253	0.030	Figure 22	HG 85-7 ST8711D-5
	BACB30DX5 BACB30DY5 NAS1465 NAS1475 (Steel)	NAS1080R05 (Steel) NAS1080-05 (2024-T4) NAS1080MG05 (Monel)	0.234 0.161	0.136	0.253	0.037	Figure 22	HG 85-12 ST8711S-5
	BACB30HD5 BACB30HD5A BACB30HC5	NAS1080-05 (2024-T4)	0.260 0.168	N/A	0.240	0.0885	Figure 23	HG 34D-9A
	BACB30GS5 BACB30GR5	NAS1080D05 (6061-T7)	0.260 0.168	N/A	0.240	0.0885	Figure 23	HG 34D-9A
0.1900	BACB30UA6 BACB30UB6 (Titanium)	BACC30BF6 (2024-T42)	0.406 0.271	0.160	0.285	0.095	Figure 22	HG 100-06 ST8707-A-06
	BACB30GP6 BACB30GQ6 NAS1526 NAS1536 (7075-T6)	NAS1080D06 (6061-T7)	0.280 0.208	0.164	0.303	0.039	Figure 22	HG 85-10 ST8707-A-06
	BACB30DX6	NAS1080R06	0.280 0.208	0.164	0.303	0.039	Figure 22	HG 85-10 ST8707-A-06

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)

TABLE XXVI - STANDARD LOCKBOLT INSPECTION DIMENSION (Continued)

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y	Z	R	T	FIG. NO.	GAGE NO.
0.1900	BACB30DY6 BACB30GX6 NAS1466 (Steel)	(Steel) BACC30Q6 NAS1080-06 (2024-T4)	0.280 0.208	0.164	0.303	0.039	Figure 22	
	BACB30HD6 BACB30HD6A BACB30HC6	NAS1080-06 (2024-T4)	0.289 0.218	N/A	0.297	0.0735	Figure 23	HG 34D-2 ST8711H-6
	BACB30HC6 BACB30HD6	BACC30L6 (Monel)	0.289 0.237	N/A	0.289	0.075	Figure 23	ST8707-6
	BACB30GS6 BACB30GR6	NAS1080D06 (6061-T7)	0.323 0.238	N/A N/A	0.297	0.0675	Figure 23	HG 34D-1
	BACB30GW6 BACB30GY6 BACB30LD6	BACC30K6 (2024-T4) NAS1080E06 (Steel)	0.264 0.191	0.164	N/A	N/A	Figure 22	HG 77-14A
	BACB30UC6 BACB30UD6 (Titanium)	BACC30K6 (2024-T4)	0.343 0.232	N/A	0.285	0.123	Figure 23	HG 120-6 ST8728-B-06
	BACB30UC6( )A BACB30UD6( )A (Titanium)	BACC30K6 (2024-T4)	0.355 0.244	N/A	0.285	0.123	Figure 23	HG 75-2 ST8728-1
	BACB30TY6 BACB30TZ6	BACC30BE6 (2024-T73)	0.343 0.228	0.160	0.285	0.094	Figure 22	HG 99-06 ST8728-B-06
	BACB30AR6 BACB30AT6	BACC30K6 (2024-T4) BACC30L6 (Monel)	0.266 0.194	N/A	0.2785	0.115	Figure 23	HG 75-1 ST8727-1
	BACB30UC6 BACB30UD6 (Titanium)	BACC30BE6 (2024-T73)	0.343 0.232	N/A	0.285	0.094	Figure 23	HG 106-06 ST8728-B-06
	BACB30UC6( )A BACB30UD6( )A (Titanium)	BACC30BE6 (2024-T73)	0.355 0.244	NA	0.285	0.094	Figure 23	HG 125-06
0.2500	BACB30UA8 BACB30UB8 (Titanium)	BACC30BF8 (2024-T42)	0.487 0.349	0.214	0.380	0.120	Figure 22	HG 100-08 ST8707-A-08
	BACB30GP-8 BACB30GQ-8 NAS1528 NAS1538 (7075-T6)	NAS1080D08 (6061-T7)	0.374 0.295	0.224	0.400	0.038	Figure 22	HG85-2 ST8711DS-8

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)

TABLE XXVI - STANDARD LOCKBOLT INSPECTION DIMENSION (Continued)

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y	Z	R	T	FIG. NO.	GAGE NO.
0.2500	BACB30DX8 BACB30DY8 BACB30GX8 NAS1468 NAS1478 (Steel)	NAS1080R08 (Steel) BACC30Q8 NAS1080-08 (2024-T4)	0.374 0.295	0.224	0.400	0.038	Figure 22	HG85-2 ST8711DS-8
	BACB30HD8 BACB30HD8A BACB30HC8	NAS1080-08 (2024-T4)	0.379 0.301	N/A	0.394	0.0995	Figure 23	HG34D-4 ST8707-8
	BACB30HD8 BACB30HC8	BACC30L8 (Monel)	0.372 0.300	N/A	0.378	0.096	Figure 23	ST8711E
	BACB30GS8 BACB30GR8	NAS1080D08 (6061-T7)	0.400 0.307	N/A	0.394	0.0735	Figure 23	HG34D-3
	BACB30GW8 BACB30GY8 BACB30LD8	BACC30K8 (2024-T4) NAS1080E08 (Steel)	0.243 0.316	0.224	N/A	N/A	Figure 22	HG77-13A
	BACB30TY-8 BACB30TZ-8 (Titanium)	BACC30BE8 (2024-T73)	0.411 0.289	0.214	0.380	0.121	Figure 22	HG99-08 ST8728-B-08
	BACB30UC8 BACB30UD8 (Titanium)	BACC30K8 (2024-T4)	0.410 0.302	N/A	0.380	0.147	Figure 23	HG120-8 ST8728-B-08
	BACB30UC8( )A BACB30UD8( )A (Titanium)	BACC30K8 (2024-T4)	0.430 0.305	N/A	0.380	0.147	Figure 23	HG132-8W ST8728-B-08
	BACB30AR8 BACB30AT8	NAS1080E08 (Steel)	0.318 0.246	N/A	0.3695	0.140	Figure 23	HG75-2 ST8728-2
	BACB30UC8 BACB30UD8 (Titanium)	BACC30BE8 (2024-T73)	0.410 0.302	N/A	0.380	0.089	Figure 23	HG106-08 ST8728-B-08
	BACB30UC8( )A BACB30UD8( )A (Titanium)	BACC30BE8 (2024-T73)	0.430 0.305	N/A	0.380	0.089	Figure 23	HG125-08
0.3125	BACB30UA10 BACB30UB10 (Titanium)	BACC30BF10 (2024-T42)	0.591 0.452	0.279	0.462	0.184	Figure 22	HG100-10 ST8707-A-10
	BACB30GP10 BACB30GQ10 NAS1530	NAS1080D10 (6061-T7)	0.459 0.339	0.268	0.486	0.083	Figure 22	HG85-3 ST8711D-10

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)

TABLE XXVI - STANDARD LOCKBOLT INSPECTION DIMENSION (Continued)

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y	Z	R	T	FIG. NO.	GAGE NO.
0.3125	NAS1540 (7075-T6)	NAS1080D10 (6061-T7)	0.459 0.339	0.268	0.486	0.083	Figure 22	HG85-3 ST8711D-10
	BACB30DX10 BACB30DY10	NAS1080P10 (2024-T4)	0.492 0.404	0.268	0.473	0.110	Figure 22	HG85-8 ST8711F-10
	BACB30GX10 NAS1470 NAS1480 (Steel)	NAS1080R10 (Steel)	0.492 0.404	0.268	0.473	0.110	Figure 22	HG85-8 ST8711F-10
	BACB30HD10 BACB30HD10A BACB30HC10	NAS1080-10 (2024-T4)	0.450 0.384	N/A	0.486	0.1065	Figure 23	HG34D-6 ST8707-10
	BACB30GS10 BACB30GR10	NAS1080D10 (6061-T7)	0.480 0.399	N/A	0.499	0.0645	Figure 23	HG34D-5
	BACB30GW10 BACB30GY10 BACB30LD10	BACC30K10 (2024-T4) NAS1080E10 (Steel)	0.341 0.268	0.268	N/A	N/A	Figure 25	HG76-3 ST8729-3
	BACB30TY10 BACB30TZ10 (Titanium)	BACC30BE10 (2024-T73)	0.496 0.385	0.278	0.462	0.150	Figure 22	HG99-10 ST8728-B-10
	BACB30AR-10 BACB30AT-10	NAS1080E10 (Steel)	0.337 0.265	N/A	0.471	0.130	Figure 23	HG75-3 ST8728-3
	BACB30UC10 BACB30UD10 (Titanium)	BACC30BE10 (2024-T73)	0.498 0.394	N/A	0.486	0.150	Figure 23	HG106-10 ST8728-B-10
	BACB30UC10 BACB30UD10 (Titanium)	NAS1080K10 (2024-T4)	0.498 0.394	N/A	0.462	0.157	Figure 23	HG120-10 ST8728-B-10
	BACB30UC10(10)A BACB30UD10( )A (Titanium)	NAS1080K10 (2024-T4)	0.523 0.412	N/A	0.486	0.146	Figure 23	HG132-10
0.3750	BACB30UA12 BACB30UB12 (Titanium)	BACC30BF12 (2024-T42)	0.642 0.503	0.341	0.572	0.156	Figure 22	HG100-12 ST8707-A-12
	BACB30GP12 BACB30GQ12 NAS1532 NAS1542 (7075-T6)	NAS1080D12 (6061-T7)	0.549 0.411	0.339	0.602	0.062	Figure 22	HG85-4 ST8711D-12
	BACB30DX12	NAS1080P12	0.604 0.507	0.339	0.576	0.120	Figure 22	HG85-9

Copyright © Boeing. All rights reserved. See Cover Page.

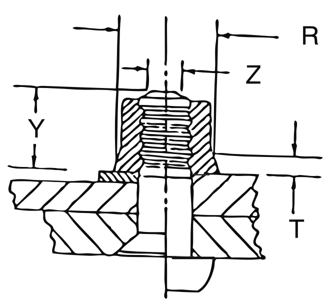
11.3.2 LOCKBOLTS (Continued)

TABLE XXVI - STANDARD LOCKBOLT INSPECTION DIMENSION (Continued)

NOMINAL DIAMETER INCH	LOCKBOLT PART NO.	COLLAR PART NO.	Y	Z	R	T	FIG. NO.	GAGE NO.
0.3750	BACB30DY12 BACB30GX12 NAS1472 NAS1482 (Steel)	(2024-T4) NAS1080R12 (Steel)	0.604 0.507	0.339	0.576	0.120	Figure 22	ST8711F-12
	BACB30HD12 BACB30HD12A BACB30HC12	NAS1080-12 (6061-T7)	0.546 0.480	N/A	0.589	0.1565	Figure 23	HG34D-8 ST8707-12
	BACB30GS12 BACB30GR12	NAS1080D12 (6061-T7)	0.583 0.496	N/A	0.589	0.1195	Figure 23	HG34D-7
	BACB30GW12 BACB30GY12 BACB30LD12	BACC30K12 (2024-T4) NAS1080E12 (Steel)	0.390 0.319	0.339	N/A	N/A	Figure 24	HG77-17A
	BACB30TY12 BACB30TZ12 (Titanium)	BACC30BE12 (2024T73)	0.531 0.420	0.339	0.572	0.146	Figure 22	HG99-12 ST8728-B-12
	BACB30AR12 BACB30AT12	NAS1080E12 (Steel)	0.380 0.308	N/A	0.5645	0.160	Figure 23	HG75-4 ST8728-4
	BACB30UC12 BACB30UD12 (Titanium)	BACC30BE12 (2024-T73)	0.531 0.420	N/A	0.572	0.146	Figure 23	HG99-12 ST8728-B-12

FL 1 Gage numbers starting with HG are produced by Huck Fasteners.

NOTE: See Figure 25, Figure 26 or Figure 29 for use of gages



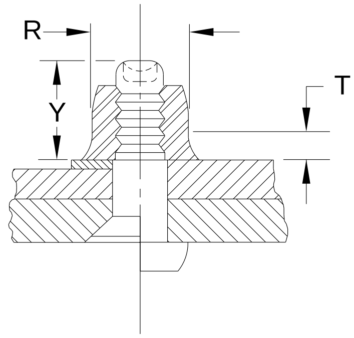
R – Swaged collar Ref. Dia.
T – Maximum height of R above sheet or washer
Y – Pin Protrusion – Height of Z Diameter above Sheet or Washer
Z – Ref. Dia Locates Measuring Point for Y

FIGURE 22 - INSPECTION OF PULL-TYPE LOCKBOLTS AND COLLARS

NOTE: See Figure 27 or Figure 29 for use of gages

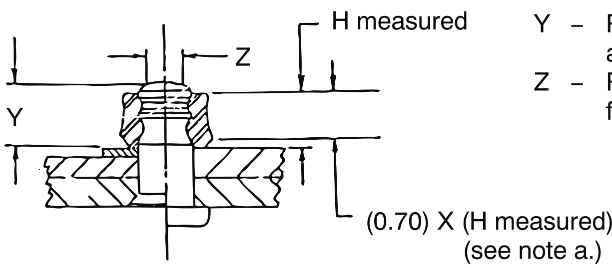
Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)



R - Swaged Collar Ref. Dia.
T - Maximum height of R above Sheet or Washer
Y - Pin Protrusion - Height of the Bolt above Sheet or Washer

FIGURE 23 - INSPECTION OF STUMP-TYPE LOCKBOLTS AND COLLARS



Y – Pin Protrusion – Height of Z Diameter above Sheet or Washer
Z – Ref. Dia – locates measuring point for Y

NOTE: a. Reversible shear collars (example: BACC30K) shall be acceptable, as pulled, when swaging is evident over 70 percent minimum of the height of the installed collar.

b. See Figure 28 for use of gages

FIGURE 24 - INSPECTION OF PULL-TYPE LOCKBOLTS AND REVERSIBLE COLLARS

1. Point touch pin
2. Legs of gage clear sheet line
3. Pin is acceptable

Accept

1. Points do not touch end of pin
2. Legs of gage touch sheet line
3. Pin grip length is too short

Reject

1. Points A do not touch pin
2. Legs of gage touch sheet line
3. Points B do not touch collar
4. Pin grip and collar swage is acceptable

Accept

1. Points A touch pin
2. Legs of gage clear sheet line or Points B touch collar
3. Pin grip is too long, collar not swaged enough

Reject

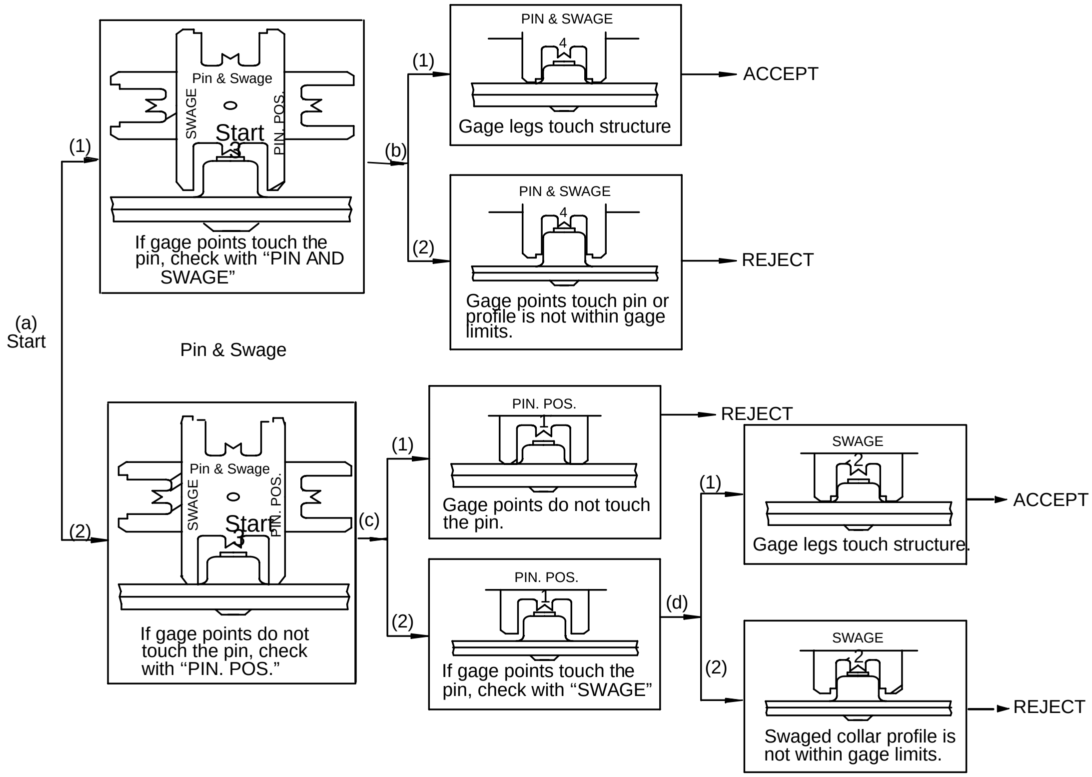
Note: Both ends of the gage shall ACCEPT

(Except see Figure 26 for HG 85-10, HG-100-( ) and HG 99-( ))

FIGURE 25 - USE OF GAGES

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)



Step a Start (Used to check lockbolt protrusion height):

1. If gage points touch the pin, the protrusion height is between nominal (0.225) and maximum (0.280). Check for acceptance with PIN AND SWAGE (Step b).

2. If gage points do not touch the pin, the protrusion is between nominal (0.225) and minimum (0.208). Check for minimum height with PIN POS. (Step c).

Step b Pin and swage, (Used to check minimum protrusion height and swaged collar profile for nominal to maximum protrusion height installation):

1. Accept the installation if gage legs touch structure.

2. Reject if gage points touch the pin (protrusion too high) or if profile is not within gage limits.

Step c Pin Position, (Used to check minimum protrusion height):

1. Reject if gage points do not touch the pin (protrusion height shorter than minimum).

2. If gage points touch the pin, check swage profile with Swage (Step d).

Step d Swage (Used to check swaged collar profile for protrusion height, nominal minimum installation):

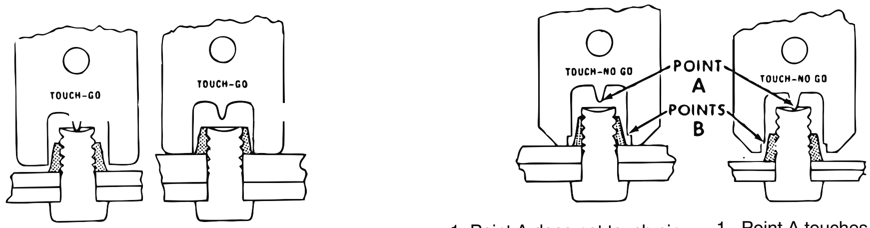
1. Accept the installation if gage legs touch the structure.

2. Reject if swaged collar profile is not within gage limits.

FIGURE 26 - USE OF GAGE

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)



1. Point touches pin
2. Legs of gage clear sheet line
3. Pin is acceptable
4. Gage pin with touch–no go end

Accept

1. Points does not touch pin
2. Legs of gage touch sheet line
3. Pin grip is too short

Reject

1. Point A does not touch pin
2. Legs of gage touch sheet line
3. Points B do not touch collar
4. Pin grip and collar swage is acceptable
5. Gage pin with touch–go end

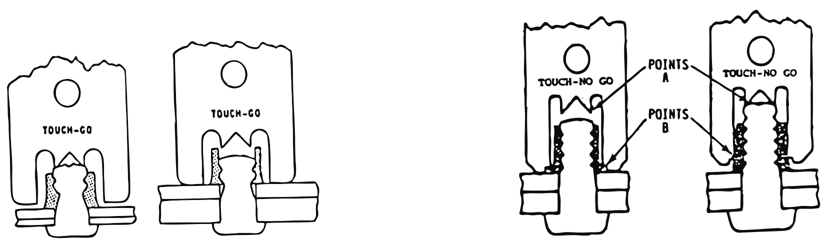
Accept

1. Point A touches pill
2. Legs of gage clear sheet line
3. Points B touch collar
4. Pin grip is too long, collar not swaged enough

Reject

Both ends of the gage shall ACCEPT

FIGURE 27 - USE OF GAGES



1. Points touch pin
2. Legs of gage clear sheet line
3. Pin is acceptable
4 Gage pin with touch–go end

ACCEPT

1. Points do not touch end of pin
2. Legs of gage touch sheet line
3. Pin grip is too short

REJECT

1. Points A does not touch pin
2. Legs of gage touch sheet line
3. Points B do not touch collar
4. Pin grip and collar swage is acceptable
5. Gage pin with touch–go end

ACCEPT

1. Point A touch pin
2. Legs of gage clear sheet line
3. Points B touch collar
4. Pin grip is too long, collar not swaged enough

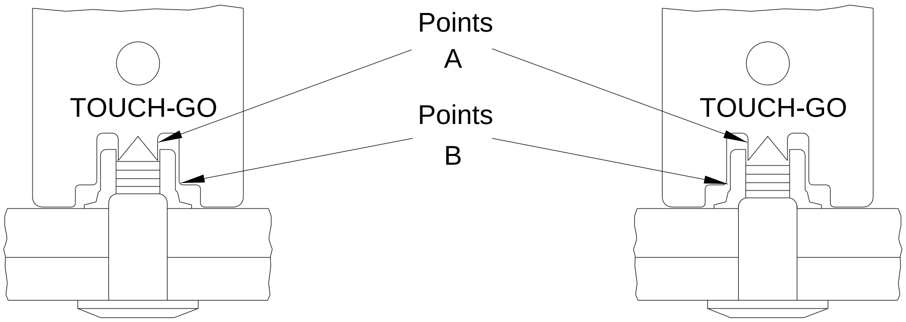
REJECT

Both ends of the gage shall accept

FIGURE 28 - USE OF GAGES

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.2 LOCKBOLTS (Continued)

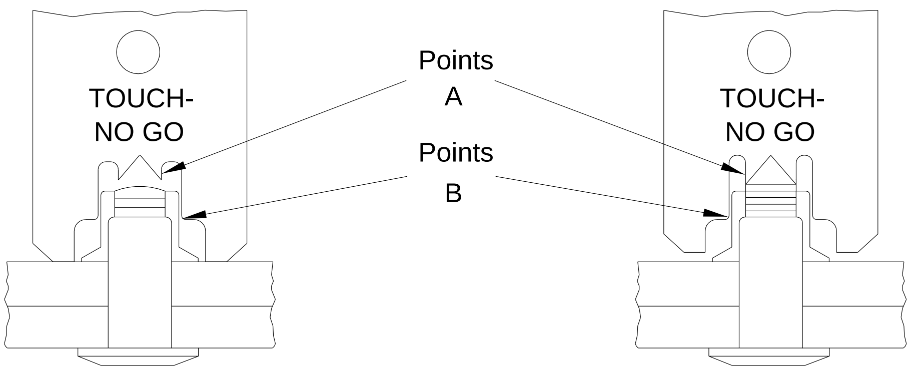


Points A touch pin. Grip length is not too short. Points B do not touch - collar is swaged enough.

ACCEPT **FL 1**

Points A do not touch pin. Grip length is too short and/or Points B touch collar - collar not swaged enough.

REJECT



Points A do not touch pin. Grip length is not too long. Points B do not touch collar-collar is swaged enough.

ACCEPT **FL 1**

Points A touch pin. Grip length is too long and/or Point B touch collar-collar not swaged enough.

REJECT

FL 1 Both ends of the gage must ACCEPT

FIGURE 29 - USE OF GAGES

11.3.3 HEX DRIVE BOLTS

a. The pin protrusion after installation shall be within the limits indicated in Table XXVII, Table XXVIII and Figure 30.

b. Pin protrusion shall be checked from the structure surface when counterbore substitution washers are used or when no grip adjustment washer is present, and from the collar/nut side surface of grip adjustment washer(s) where such washers are used (See Figure 13).

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.3 HEX DRIVE BOLTS (Continued)

c. Pin protrusion shall be measured from the top of self-aligning washers (for example, BACW10CA) if used under collar or nut.

d. If self aligning nuts, (for example, BACN10MT) are used on sloping surfaces. Pin protrusion 'P' shall be measured as shown in Figure 31.

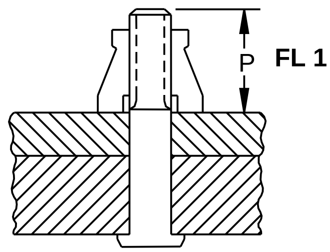


FIGURE 30 - HEX DRIVE BOLT PIN PROTRUSION

FL 1 Protrusions apply after installation of collar or nut.

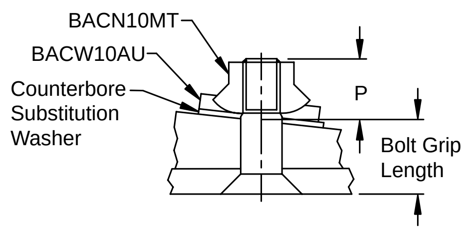


FIGURE 31 - THREAD PROTRUSION ON SLOPING SURFACE

TABLE XXVII - PIN PROTRUSION-HEX DRIVE BOLTS (EXCEPT NICKEL ALLOY 718 HEX DRIVE BOLTS-BACB30YK, BACB30YL, BACB30YM, BACB30YN, HLT 420, HLT 421, HLT 422, HLT 423)

FASTENER		STANDARD WEIGHT (INCH)							LIGHTWEIGHT (INCH)		
		NOMINAL, 0.0156 AND 0,0312 OVERSIZE, INCH		GAGE NO.	0.0468 OVERSIZE				NOMINAL		GAGE NO.
1ST DASH NO.	NOMINAL THREAD	SHEAR AND TENSION		ST8712 or OM2-1522-B FL 1	SHEAR		TENSION		SHEAR		OM2-1522HST-B or ST8712B FL 2
		P MIN	P MAX		P MIN	P MAX	P MIN	P MAX	P MIN	P MAX	
-5	0.1640-32	0.302	0.389	-5 (5/32)	0.333	0.414	0.315	0.397	0.270	0.352	-05 (5/32)
-6	0.1900-32	0.315	0.402	-6 (3/16)	0.346	0.427	0.350	0.432	0.280	0.362	-06 (3/16)
-8	0.2500-28	0.385	0.472	-8 (1/4)	0.416	0.497	0.420	0.502	0.310	0.392	-08 (1/4)
-10	0.3125-24	0.490	0.577	-10 (5/16)	0.521	0.602	0.520	0.602	0.370	0.452	-10 (5/16)
-12	0.3750-24	0.535	0.622	-12 (3/8)	0.566	0.647	0.565	0.647	0.410	0.492	-12 (3/8)
-14	0.4375-20	0.625	0.707	-14	0.656	0.737	0.650	0.732	0.475	0.557	- - -
-16	0.5000-20	0.675	0.757	-16	0.706	0.787	0.700	0.782	0.515	0.597	- - -

Copyright © Boeing. All rights reserved. See Cover Page.

11.3.3 HEX DRIVE BOLTS (Continued)

TABLE XXVII - PIN PROTRUSION-HEX DRIVE BOLTS (EXCEPT NICKEL ALLOY 718 HEX DRIVE BOLTS-BACB30YK, BACB30YL, BACB30YM, BACB30YN, HLT 420, HLT 421, HLT 422, HLT 423)
(Continued)

FASTENER		STANDARD WEIGHT (INCH)							LIGHTWEIGHT (INCH)		
		NOMINAL, 0.0156 AND 0,0312 OVERSIZE, INCH		GAGE NO.	0.0468 OVERSIZE				NOMINAL		GAGE NO.
1ST DASH NO.	NOMINAL THREAD	SHEAR AND TENSION		ST8712 or OM2-1522-B FL 1	SHEAR		TENSION		SHEAR		OM2-1522HST-B or ST8712B FL 2
		P MIN	P MAX		P MIN	P MAX	P MIN	P MAX	P MIN	P MAX	
-18	0.5625-18	0.760	0.842	-18	0.791	0.872	0.790	0.872	0.590	0.672	- - -
-20	0.6250-18	0.815	0.897	-20	0.846	0.927	0.865	0.947	0.630	0.712	- - -
-24	0.7500-16	1.040	1.122	-24	1.071	1.152	1.090	1.172	- - -	- - -	- - -
-28	0.8750-14	1.200	1.282	-28	1.231	1.312	- - -	- - -	- - -	- - -	- - -
-32	1.0000-12	1.380	1.462	-32	1.411	1.492	- - -	- - -	- - -	- - -	- - -

FL 1 OM2-1522-B gages are available from Omega Technologies.

FL 2 OM2-1522HST-B gages are available from Omega Technologies.

TABLE XXVIII - PIN PROTRUSION-HEX DRIVE BOLTS FOR NICKEL ALLOY 718 HEX DRIVE BOLTS - BACB30YK, BACB30YL, BACB30YM, BACB30YN, HLT 420, HLT 421, HLT 422, HLT 423

FASTENER		NOMINAL		0.0156 AND 0.0312 OVERSIZE	
		SHEAR AND TENSION		SHEAR AND TENSION	
FIRST DASH NUMBER	NOMINAL THREAD	P MIN (MINIMUM PROTRUSION)	P MAX (MAXIMUM PROTRUSION)	P MIN (MINIMUM PROTRUSION)	P MAX (MAXIMUM PROTRUSION)
-5	0.1640-32	0.302	0.384	- - -	- - -
-6	0.1900-32	0.315	0.397	0.350	0.432
-8	0.2500-28	0.385	0.467	0.425	0.507
-10	0.3125-24	0.490	0.572	0.535	0.617
-12	0.3750-24	0.535	0.617	0.580	0.662
-14	0.4375-20	0.625	0.707	0.680	0.762
-16	0.5000-20	0.675	0.757	0.730	0.812
-18	0.5625-18	0.760	0.842	0.815	0.897
-20	0.6250-18	0.815	0.897	0.880	0.962

11.3.4 HEX-DRIVE BOLTS WITH MACHINE SWAGED LOCKBOLT COLLARS

a. When collars are installed against a sloping surface, the pin protrusion and collar swage height shall be checked with the plane of the protrusion gage oriented perpendicular to the slope of the surface.

Copyright © Boeing. All rights reserved. See Cover Page.

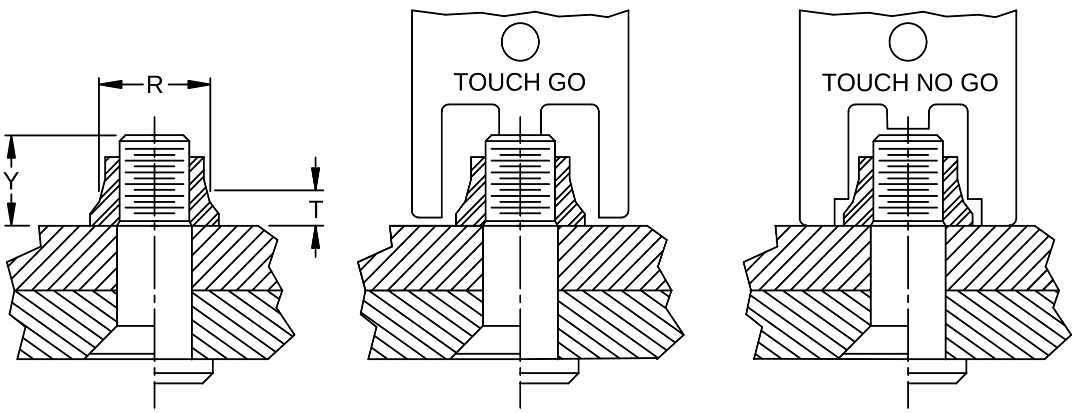
11.3.4 HEX-DRIVE BOLTS WITH MACHINE SWAGED LOCKBOLT COLLARS (Continued)

b. Slivers of collar material visible above the bolt point are acceptable.

c. Lightweight Hi-Loks with swaged lockbolt collars shall be installed in accordance with Table XXIX and Figure 32. Swage height and pin protrusion shall be inspected using the applicable GO-NO-GO gage as shown in Figure 32.

TABLE XXIX - INSTALLATION REQUIREMENTS FOR HEX DRIVE WITH SWAGED COLLARS

NOMINAL DIAMETER INCH	PIN PROTRUSION Y		GAGE DIA R ± 0.0005	SWAGE HEIGHT T MAX
	MIN	MAX		
0.190	0.280	0.362	0.2620	0.102
0.250	0.310	0.392	0.3440	0.107
0.313	0.370	0.452	0.4360	0.102



1. Condition shown is acceptable.
2. If gage touches sheet line, pin protrusion is not acceptable.

1. Condition shown is acceptable.
2. If gage touches pin, pin protrusion is not acceptable.
3. If gage touches collar, swage height is not acceptable.

FIGURE 32 - INSPECTION OF HI-LOKS WITH SWAGED COLLARS

11.4 LIMITATIONS FOR BOLTS INSTALLED BY PULL-IN PROCESS (IN ACCORDANCE WITH SECTION 8.2.8)

a. This process is limited to BACB30XT( )K, BACB30VM( )K and BACB30VN( )K lockbolts.

b. Bolt to hole interference and bolt grip length not to exceed values in Table XXX.

c. Limited to 2XXX and 7XXX series aluminum structure only.

Copyright © Boeing. All rights reserved. See Cover Page.

11.4 LIMITATIONS FOR BOLTS INSTALLED BY PULL-IN PROCESS (IN ACCORDANCE WITH SECTION 8.2.8) (Continued)

TABLE XXX - BOLT INTERFERENCE LIMITS

MAXIMUM FASTENER DIAMETER, INCH	MAX. BOLT TO HOLE INTERFERENCE, INCH	MAX. BOLT GRIP LENGTH
3/8	0.004	-16
5/16	0.003	-14
1/4	0.003	-11
3/16	0.003	-8

12 TEST METHODS

Not applicable to this specification.

13 QUALITY CONTROL

Not applicable to this specification.

Copyright © Boeing. All rights reserved. See Cover Page.